Airbag-Based Protective System for UAV Medical Payloads

Design and Simulation in LS-DYNA

Annamaria Iannuzzo



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by

Annamaria Iannuzzo

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Preface

A preface...

Annamaria Iannuzzo
Delft, October 2026

Summary

A summary...

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Nomenclature

Abbreviations

Abbreviation	Definition
ABS	Acrylonitrile butadiene styrene
AOU	*Azienda Ospedaliero-Universitaria* (University Hospital Trust)
ASL	*Azienda Sanitaria Locale* (Local Health Authority)
ASST	*Azienda Socio-Sanitaria Territoriale* (Territorial Social and Health Authority)
ConOps	Concept of Operations
CTO	*Centro Traumatologico Ortopedico* (Orthopaedic Trauma Centre, Turin)
DIMEAS	*Dipartimento di Ingegneria Meccanica e Aerospaziale* (Department of Mechanical and Aerospace Engineering, Politecnico di Torino)
DoF	Degree(s) of freedom
DTA	*Distretto Tecnologico Aerospaziale* (Apulian Aerospace Technology Cluster)
EASA	European Union Aviation Safety Agency
ENAC	*Ente Nazionale per l'Aviazione Civile* (Italian Civil Aviation Authority)
ENAV	*Ente Nazionale di Assistenza al Volo* (Italian Air Navigation Service Provider)
EU	European Union
eVTOL	Electric vertical take-off and landing
FEM	Finite Element Method
H2020	Horizon 2020 (EU research and innovation programme)
JARUS	Joint Authorities for Rulemaking on Unmanned Systems
MTOM	Maximum take-off mass
PET	Polyethylene terephthalate
PET-G	Polyethylene terephthalate glycol-modified
PIC4SeR	PoliTo Interdepartmental Centre for Service Robotics
PU	Polyurethane
RRT	Rapidly-exploring Random Tree (RRT*: asymptotically optimal variant)
SIS 118	*Società Italiana Sistema 118* (Italian Society of the 118 Emergency Medical System) [verify]
SORA	Specific Operations Risk Assessment
SSN	*Sistema Sanitario Nazionale* (Italian National Healthcare System)
SSR	*Sistema Sanitario Regionale* (Regional Healthcare Service)
UAV	Unmanned aerial vehicle
UN3373	UN number for Biological Substance, Category B (transport packaging standard)
VTOL	Vertical take-off and landing

Symbols

Symbol	Definition	Unit
Latin symbols		
A_t	Airbag–ground footprint area at time t	$[\mathrm{m}^2]$
A_{vent}	Effective vent area	$[\mathrm{m}^2]$
a	Payload acceleration	$[\mathrm{m\,s}^{-2}]$
a_{allow}	Allowable payload acceleration	$[\mathrm{m\,s}^{-2}]$
a_{lim}	Design acceleration limit, a_{allow}/SF	$[\mathrm{m\,s}^{-2}]$

continued on next page

Symbol	Definition	Unit
D_0	Initial airbag diameter (characteristic dimension)	[m]
D_t	Airbag diameter at time t	[m]
F_{fail}	Phial failure force under axial compression	[N]
F_n	Normal ground reaction force on the airbag	[N]
FT_{fail}	Phial brittle failure threshold	[verify]
g	Gravitational acceleration	[$\text{m}\,\text{s}^{-2}$]
h	Stroke, i.e. payload travel while compressing the airbag	[m]
k	Airbag shape coefficient (sphere: 0.5; cylinder: 1) [verify]	[–]
L_0	Initial airbag length (cylindrical airbag)	[m]
M, M_{p}	Payload mass	[kg]
m	Gas mass inside the airbag	[kg]
$\dot{m}$	Vented gas mass flow rate	[$\text{kg}\,\text{s}^{-1}$]
$\mathbf{n}$	Unit vector along the impact (stroke) direction	[–]
n_{ply}	Number of fabric plies	[–]
P_0	Initial inflation pressure	[Pa]
P_{amb}, P_{atm}	Ambient (atmospheric) pressure	[Pa]
P_{bag}	Airbag internal pressure	[Pa]
P_{burst}	Airbag burst pressure	[Pa]
P_{open}	Vent opening threshold pressure	[Pa]
PEN	Penalty weight on constraint violations	[–]
R	Specific gas constant	[$\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$]
SF	Safety factor on the allowable acceleration	[–]
SF_{burst}	Design factor on membrane tension	[–]
T	Gas temperature	[K]
T_0	Initial gas temperature	[K]
t	Time	[s]
t_0	Initial time (onset of impact)	[s]
t_{fall}	Free-fall time	[s]
u_0	Initial vertical impact velocity	[$\text{m}\,\text{s}^{-1}$]
u_t	Payload velocity at time t	[$\text{m}\,\text{s}^{-1}$]
$u_{n,0}$	Initial impact velocity along $\mathbf{n}$	[$\text{m}\,\text{s}^{-1}$]
$u_{\text{res,max}}$	Maximum allowable residual velocity	[$\text{m}\,\text{s}^{-1}$]
$u_{x,0}$	Initial horizontal impact velocity	[$\text{m}\,\text{s}^{-1}$]
V_0	Initial airbag volume	[m^3]
V_t	Airbag volume at time t	[m^3]
V_x, V_y	Horizontal and vertical impact velocity	[$\text{m}\,\text{s}^{-1}$]
v	Impact velocity	[$\text{m}\,\text{s}^{-1}$]
Greek symbols		
γ	Specific heat ratio	[–]
Δm	Gas mass vented in one timestep	[kg]
Δt	Timestep	[s]
η_{seam}	Seam efficiency (fraction of fabric strength retained at seams)	[–]
θ	Impact angle, measured from the vertical	[°]
ρ_t	Gas density at time t	[$\text{kg}\,\text{m}^{-3}$]
φ_{vent}	Vent open-area fraction	[–]

1

Introduction

1.1. Motivation

The present MSc Thesis in Aerospace Structures and Materials falls under the category of *Engineering Design*, in alternative to the *Scientific Research*-type projects. In *design* projects, it is required to attempt the process of innovation not through the discovery and exploration of a research niche, but rather by applying pre-existing scientific knowledge to enable the concrete realisation of a technical design. In the case of the present thesis, the overall objective consists in proposing a *design workflow* and validating it through testing and simulation. If adopted in the preliminary design phase of a drone-based delivery mission, this workflow yields an optimised airbag-type device tailored to a specific failure-and-drop application, in line with the requirements set by the project commissioners. Therefore, the successful completion of this thesis is achieved by:

1. Proposing an application-specific design workflow that fits within the broader context of the ongoing research effort;
2. Testing the workflow, both experimentally and through simulation, to realistically inform the device's design.

The design workflow (informally called *inner design loop* throughout this thesis) is inteded as a first attempt to bridge the gap between the long-term, broadly-defined objective of constructing a large-scale drone network (also called *outer (design) loop*) over the region of Piedmont, Italy, and the previously completed research work, consisting in a series of logistic, legislative, and generally high-level evaluations of possible **drone-based missions** with the objective to **deliver medical supplies in emergency scenarios**. Such work was conducted *Politecnico di Torino*, in collaboration with the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex and automotive consulting company *FEV Italia S.r.l.*, three of the main industrial and research stakeholders working towards helping hospitals in the transition from road to air transport to enhance the logistics of the *Sistema Sanitario Nazionale* (*SSN*, the Italian National Healthcare System) both at an intra- and inter-hospital level.

This MSc Thesis Project is, therefore, a multidisciplinary project where the knowledge and techniques of Structural Engineering are put to the service of the medical field. The airbag-like device resulting from the design process should be able to protect the free-falling medical payload in the event of a drone failure and its subsequent detachment from the latter. The device aims to bring the impact accelerations within the allowable limits of the payload container and of the medical payload being transported, in compliance with the safety regulations regarding dispersion of chemical substances into the environment. However, contrary to most aeronautical projects which do not distinguish between the integrity of the aerial vehicle and the safety of its passengers, this project adopts a "payload-first" approach where the protection of the highly valuable medical cargo is prioritised over the integrity of the drone carrying it. Distinguishing between the safety aspects of the unmanned aerial vehicle (UAV) and its payload implies a certain facilitation in the certification process of the UAV, as well as for the modular implementation of the safety device on a variety of UAVs serving in different, specific delivery

missions.

1.2. Project framing

1.2.1. Care centralisation and territorial inequity in the Italian healthcare system

The idea of transforming healthcare delivery networks from road- to air-transport-based systems stems from the consensus within the Italian medical field, consolidated during the public health emergency of the COVID-19 pandemic (2020–2022), that the current ambulance-based architecture of the *SSN*'s logistic infrastructure (described in detail[1] in [1]) cannot accommodate the healthcare system's needs and transitions. The two main - and competing - factors are:

1. **Care centralisation**: [2] describes care centralisation as the ongoing process of concentrating medical expertise and services in key geographical areas, e.g. provincial and/or regional capitals, to meet the demands of high-volume care. The downside of care centralisation is that at a regional level, it induces "distance-related inequity, particularly for rural and peripheral populations", especially heightened in emergency scenarios.
2. **Unhomogeneous distribution of emergency services** : [3] affirms that, in contrast with the centralisation manouver described above, emergency resources are still unhomogeneously distributed over the regional territories, with services overloaded in densely populated areas and underused in low-density ones, resulting in inefficient overall utilisation.

In conclusion, [4] builds on these two points by affirming the importance of establishing a generally more efficient strategy for healthcare management at all levels, and incentivises the development of technologies suitable for upgrading, among the several aspects of the *SSN*, the logistics of the Italian healthcare supply chain.

1.2.2. Prior research and positioning of the present thesis

Several industry-research partnerships aiming to modernise transportation means within the Italian *SSN* have ventured into the globally booming market of unmanned aerial vehicles, where remotely-piloted or fully autonomous drones are being used ever more often to carry medical payloads in a plethora of scenarios: intra- vs. inter-hospital delivery [5], ordinary logistics vs. emergency missions and humanitarian applications[6], experimental/direct flight between strategic poles [7] vs. a fully-operational, inter-hospital, hub-based UAV network covering national territory (the *Zipline* case study[8]).

Following the global trend to answer a local need, in the specific case of the Piedmont region in north-west Italy, where at the time of writing there exists no city- or region-wide UAV network deployed for the delivery of medical goods, many pilot programmes (further analysed in Section 2.1) are emerging, among which a partnership between the Pharmacy department of the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex (head-quartered at Molinette Hospital in Turin), a research group within the Department of Mechanical and Aerospace Engineering at *Politecnico di Torino*, and the Numerical Anaysis Team of *FEV Italia S.r.l.*, the Italian branch of the German-founded multinational company *FEV*, a global leader in research and development for the automotive industry.

The six-year-long partnership between these three institutions is the backbone of the present MSc thesis project, the most recent of a chain of research projects aiming to pursue the long-term vision of flanking, if not completely replacing, the current road-based transportation means with aerial alternatives within hospitals in Piedmont and, specifically, in the city of Turin (the region's capital), in favour of more rapid, more efficient, and safer transportation of medical supplies. Specifically, this research effort has broadly tackled the key aspects required to establish a complete and regulation-compliant UAV medical delivery ecosystem in the Piedmontese territory, resulting in five MSc theses and one conference paper presenting the following results:

- **Cavallo (2020)** [9]: Preliminary assessment of UAV platforms suitable under Italian and European regulations. Development of risk maps and definition of optimal trajectories between selected

[1] *"The type of vehicles used varies from region to region. In Piedmont it mainly consists of ambulances used only for transport plus more advanced vehicles equipped for resuscitation, replaced in some sites by mSa on a car with a doctor, nurse and rescuer driver, organized on the basis of specific needs of the territory, different in the four macro-areas of the emergency network (rEmE) in which the region was divided."*[1]

hospitals.

- **Bertelli (2021)** [10]: Detailed selection of UAV based on mission specifications derived from the medical goods to be transported and ENAC regulatory constraints. Application of the SORA (Specific Operations Risk Assessment) methodology to three feasible mission routes, yielding the safety requirements that operators must satisfy.
- **Sivieri (2021)** [11]: Identification and characterisation of thermal and vibrational loads encountered during typical UAV medical missions. Experimental testing on milk-derived sera samples to determine their resistance to UAV-induced vibrational stresses. First hypothesis of medical payload container geometry and materials, implemented in the experimental campaign.
- **Fiorentino (2021)** [12]: Development of a path-planning strategy for life-saving drug distribution in populated environments. Generation of risk maps and implementation of an RRT-based algorithm to minimise global risk and flight time.
- **Angelini (2023)** [13] and **Angelini et al.(2024)** [14]: Comparative analysis of the safety and economic efficiency of emergency medical deliveries performed by car, drone, and VTOL aircraft. Identification of optimal pharmaceutical delivery hub locations for each transport mode to maximise logistical performance.

The present thesis was carried out in collaboration with the three institutions mentioned above to advance the research effort, bridging the gap between their long-term objective and the results achieved so far. This required preliminary work aimed at formalising the project requirements at both a broad, high level and at a specific, first-step level. Having taken a legislative, logistic, market- and mission-evaluation approach until now, it was possible to delineate a scenario in which the approval of *ENAC* (the Italian Civil Aviation Authority - *Ente Nazionale per l'Aviazione Civile*) could potentially be obtained to enable this type of operation. However, as certifications are unavoidably specific to mission type, trajectory and vehicle, it appeared necessary to implement a more concrete approach rooted in *design*, whilst also adopting a "payload-first" approach that diverted the project towards a device for the *safety of the medical payload* rather, than that of the aerial vehicle. The deliberate prioritisation of this aspect is justified in Sections 2.1 and 2.2, where it is illustrated on the basis of a regional market analysis and the exploration of an ongoing paradigm shift in the design process of biomedical devices.

1.2.3. Stakeholders

Overall, the stakeholders participating in the present project had the following roles:

- **TU Delft** (Prof. Bianca Giovanardi): Primary academic supervisor overseeing the scientific quality and methodological development of the research.
- **Politecnico di Torino** (Prof. Enrico Cestino, Davide Angelini): Institutional leads and co-supervisors for the broader UAV medical-delivery research project, providing access to the experimental platforms used.
- **A.O.U. Città della Salute e della Scienza di Torino / Molinette Hospital**: Provider of medical and operational data related to pharmaceutical transport requirements and mission scenarios.
- **FEV Italia S.r.l.**: Industrial partner offering technical support for advanced modelling and simulation activities, including SolidWorks-based design and LS-DYNA impact analyses.

The interaction between stakeholders, as well as the management and communication aspect of the project, is analysed in Chapter 6 - "Project Management".

1.3. Document Outline

DISCORSIVO. Esempio preso da un'altra tesi:

The thesis is structured as follows, first, chapter 2 presents a literature review to outline the several possibilities to predict the buckling load of cylindrical shells and which techniques are available to reduce the sensitivity to imperfections. The chapter concludes by identifying the research gaps related to the buckling of cylindrical shells. From this conclusion, the research objective and scope are defined in chapter 3. The goal is to include imperfections within the displacement-based Koiter methodology and the necessary steps to achieve this are explained in chapter 4 to 7, as well as how effective this novel

modelling technique is. Finally, the concluding remarks can be found in chapter 8 and future research is presented in chapter 9.

2

Background and Research-Development Gap

In technology development, it is important to gain a clear idea of two aspects:

1. **Research aspect**: if the technology that is being developed requires the exploration of an uncharted research niche, as opposed to utilising pre-existing and consolidated knowledge for the creation of a new product;
2. **Development aspect**: whether the proposed technology targets a well-defined market with unmet customer demand and no established supply.

This chapter explores both these aspects in Sections 2.2-2.3 and 2.1, respectively. The aim is to gather sufficient background information, from market allocation of the proposed technology to concrete understanding of airbag design criteria, to depict the choices proposed in Chapter 3 regarding operative *methodology* as an answer to the Research Questions proposed at the end of this chapter.

2.1. Regional Market Analysis: medical UAV logistics in Piedmont

The Italian National Healthcare System (*SSN*) is structured hierarchically: the guidelines and regulations issued at a *national* level must be implemented locally, at a *regional* level, by each region's *Sistema Sanitario Regionale* (*SSR*, the Regional Healthcare Service). Vice versa, aviation rules follow European-level enforcement mechanisms: Following the 2019 harmonisation of European drone operations across the "open" and "specific" categories [15], operators seeking authorisation for high-risk flights are required to conduct a thorough risk analysis using the *Specific Operations Risk Assessment (SORA)* methodology developed by JARUS[1], making a detailed evaluation of potential failure scenarios while proving compliance with the demanding safety standards imposed by aviation authorities.

The discrepancy between modi operandi of the aviation and healthcare authorities has shaped the way drone-network operations have mostly been developing in the last decades in Italy: several emerging, albeit scattered, research groups, either cooperating or competing with industrial counterparts, have achieved geographically circumscribed trial deployments while being unable to scale the operation beyond testing.

A complete summary of such operations, limited to the means of investigation available for the present thesis project, can be found in Table A.2 in Appendix A.2. The table shows a clear trend: although potentially booming as several healthcare facilities are exploring the possibility of adopting drone transport and operators are proposing new drone models for them, most activities stop after the initial demonstrations, with the only exception being the region of Abruzzo where the operational phase of the drone drug-delivery service between a strategic hospital and village pharmacies was announced in July 2025 [16], [17]. In general, a pattern of fragmentation, short-lived pilots, and few transitions

[1] *Joint Authorities for Rulemaking on Unmanned Systems*

to a potentially scalable service is observed in the construction of the overall *value chain*, with many independent institutions covering equivalent roles, as shown in Table A.1 in Appendix A.1.

Table 2.1: Actors involved in drone delivery of medical goods in Piedmont, grouped by project.

Project	Role	Institution	Scope
INDOOR (2021-2024)	Promoter	Fondazione DOT	Project promoter [18]
	Funder	Centro Nazionale Trapianti	Research grant at PIC4SeR [19]
	Funder	City of Turin	Project partner [18]
	Research	PIC4SeR, Politecnico di Torino	Autonomous flight technologies for transporting biological material [20]
	Research	Università di Torino	Project partner [18]
	Technology partner	Mavtech	Technical partner; aerial surveillance and operational-support products [18]
	Operator, payload maker	ABzero (Pisa)	Supplied the drone and the Smart Capsule (sensorised container for blood, organs and drugs); handled the ENAC authorisation [19], [21], [22], [23]
	End user	AOU Città della Salute e della Scienza	Test flight between hospital sites [22]
	End user	Regional Transplant Centre Piemonte–Valle d'Aosta	Organ and biological transport [18]
	Regulator	ENAC	National civil aviation authority; project partner and flight authorisation [18]
DIMEAS feasibility studies (2020-2024)	Research	DIMEAS, Politecnico di Torino (Prof. E. Cestino)	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost–safety comparison of car, drone and eVTOL, hub siting [9], [10], [11], [12], [13], [14]
	Technology partner	FEV Italia	Industrial partner; numerical impact analysis
	End user	AOU Città della Salute e della Scienza	Pharmacy department partner
Hydrogen logistics plan	End user	AOU Città della Salute e della Scienza	Proposed hydrogen drone logistics plan [24]
	Infrastructure	Torino Caselle and Cuneo Levaldigi airports	Proposed hub and support base [24]
LastMed	Funder	City of Turin	Funded the LastMed call [25]
	Alternative (ground)	Tecnocad / Tecnomill / Bylogix	Autonomous ground rover for last-mile drug delivery [26]
Independent	Drone manufacturer	SkyEU	Autonomous high-speed eVTOL drones for first aid and critical logistics [27]
	Regulator	Regione Piemonte	Regional health authority; drone trials subject to benefit–risk analysis [28]

Adopting a region-based, rather than a nation-based, approach for market analysis is, therefore, necessary. Table 2.1 highlights stakeholders involved in Piedmont-based operations and their roles in the value chain, illustrating a fragmented scenario: Project INDOOR, launched by Fondazione DOT in 2021, culminated in a 500 m-long test flight by an autonomous UAV between two Città della Salute hospital sites, carrying medical goods in a dedicated capsule, on 1st October 2024. In this project, several other stakeholders were involved (in fact, most of the ones present in Table 2.1), among which ABzero (a spin-off from *Scuola Normale Superiore di Pisa* responsible for capsule design and handling the ENAC authorisation process for the test flight), parallel research group PIC4SeR from the Department of Electronics and Telecommunications at *Politecnico di Torino*, and *MAVTech S.r.l.* (UAV systems designer company based in Turin). Independently of the INDOOR project of 2021-2024, the startup SkyEU was founded in 2025 promising to develop eVTOL drone systems for critical logistics within, but not limited to, the healthcare sector.

In parallel, and independently of INDOOR, the research group led by Prof. Enrico Cestino at the Department of Mechanical and Aerospace Engineering (DIMEAS) of *Politecnico di Torino*, together with *A.O.U. Città della Salute e della Scienza* and *FEV Italia*, has pursued since 2020 a series of feasibility studies on the Molinette-San Giovanni Bosco route (Section 1.2.2). These covered regulatory and mission evaluation, risk mapping and path planning, payload environmental loads, and cost-safety comparisons between road, drone and eVTOL transport. No flight campaign has been carried out so far. This makes the group the only Piedmontese actor whose work so far has been entirely at the planning and assessment level.

On the other hand, the regional administration's general stance with respect to this type of operation highlights the fundamental requirement of thorough benefit-risk analyses to guarantee both the safety of the Piedmontese citizens and the preservation of the medical cargo, possibly needing to limit transportation to exceptional emergencies in mountain areas [28]. From an engineering perspective, these criticalities can be translated in the following barriers for the development of a full-scale, UAV-based delivery network:

1. **Authorisation of flights**: Operations beyond the open category are approved by compiling the SORA. Among the different classes of mitigations, *ground risk* mitigations (one of the first steps required in the SORA) reduce the consequences of the UAV's impact on the ground after a loss of control, for example by introducing a parachute. It is key for technical stakeholders to prove the reliability of their implemented ground risk mitigations to protect civilians in case of drone failure.
2. **Payload integrity and containment**: In accordance with the regional administration's concerns, current certification standard UN3373 for ground handling of Biological Substances, Category B, establishes that packages must pass a drop test at 1.2 m with no leakage from the primary container CITE STANDARD HERE, perhaps it needs a bit more detail. However, in the event of a free-fall from ordinary mission altitudes, this standard loses its significance and the potential for biological material dispersion into the environment becomes highly relevant.
3. **Unsustainability of the current business models** While costs of road and helicopter transport of biomedical goods such as organs, drugs and other critical supplies are expected to fall significantly with drones, most trials are funded by EU projects and one-time grants, with the consequence of making the project unsustainable and non-scalable.

On the basis of this analysis, several research and development gaps can be identified: SkyEU is endeavouring to enact an economically sustainable business model for large-scale UAV networks, but has a small public record; ABzero has patented capsules equipped with electronics that reliably monitor in-flight vibrations and guarantee thermal control of the medical payload, but these are only certified to the UN3373 standard, whereas in case of in-flight failure a much more conservative approach is needed; regional authorities need reassurance on the safety aspect for civilians, as a 500 m test flight on a controlled route shows that the obstacle is not drone-making, but guaranteeing safety in everyday performance rather than in single monitored missions.

In this scenario, the institutions (*Politecnico di Torino (DIMEAS), FEV Italia, AOU Città della Salute*) that commissioned the present thesis project aim to tackle, among the three barriers, barrier 2, concerning payload integrity and containment. The proposed solution is an airbag-type device that protects the payload and prevents its dispersion in the event of an impact, especially in view of transporting extremely valuable medicinals and/or biological materials, e.g. human organs for urgent transplant operations. This device is conceived as an add-on to existing capsules such as the one by ABzero or those mounted on drones such as SkyEU's (for which documented information remains unavailable at the time of writing), and is intended to help close the certification gap highlighted above, while acknowledging that it does not address the issues linked to SORA ground risk or the sustainability of the business model.

2.2. Mission-first vs. Safety-first: a design-philosophy gap

This section builds on the research-development gap presented earlier by attempting to connect its root causes to a discrepancy between design philosophies applied in the aeronautical and the biomedical sectors. The proposed design solution (the airbag-type device) is, in fact, a suggestion to solve an "aeronautical problem" implementing a "medical approach".

The issue of proving compliance to match demanding safety standards imposed by the certification authorities does not appear to be a Piedmont-only (or even an Italy-only) problem, but rather a widespread trend that affects all non-standard aeronautical projects across the field. As highlighted, for example, in [29], most UAV producers focus on optimising their aerial vehicle's design, minimising mission-related vibrational loads, thermal loads, overall mass, but also proposing ever more efficient cargo-drop mechanisms and ways to overcome geographical obstacles, all partaking in a feedback loop against aerodynamic and flight-mechanics performance metrics. This machine-centric approach appears to be the default, especially considering that UAV-based missions are, naturally, radically different from standard airline and/or general aviation operations, where passenger safety and vehicle safety are inherently aligned and where, therefore, stringent certification requirements strongly limit design freedom CITE ARTICLE HERE. A common consequence of focusing exclusively on technological potential when designing UAV-based operations is, therefore, neglecting the safety assurance aspect of the project, which is usually delegated to including a parachute-type device in the final design phases. As pointed out by [30] in their design feasibility evaluation study, the introduction of a parachute usually increases the risk of falling into a mass-increase feedback loop: the parachute increases the airframe's overall mass, which then requires a larger, thus heavier, parachute for efficient deceleration during free-fall, which again results in an excessive mass increase, and so on. This negative feedback loop results either in never reaching design convergence, or otherwise in having to accept a smaller parachute with reduced deceleration capacity, still implying a relatively high ground-impact velocity and almost certainly destructive damage to the airframe, its cargo, and the impacted areas on the ground.

The general impression given by the brief market analysis conducted earlier and the observations gathered in the present paragraph is that, perhaps, the "mission-first" approach adopted in the majority of nascent UAV-based operations is not sustainable for the projects' longevity beyond pilot flights, and that it might be necessary to embrace a paradigm shift in design approaches. Specifically, there might be potential in drawing inspiration from the biomedical field, where medical technology designers are encouraged to shift from a techno-centred to a human-centric approach to avoid the *translation gap* or *implementation gap*, i.e. the potential to introduce or enhance health inequity when trying to implement extremely advanced technology that does not take into account all necessities and limitations of human patients [31].

Just like success is measured on the patient and not on the device in the biomedical field, in the case of UAV-based medical delivery operations success should be measured on payload integrity and protection of civilians, not (only) on the capabilities of the aerial vehicle. A drone that survives an in-flight failure while sacrificing its payload and threatening irreversible human damage has undoubtedly failed its mission, as for a biomedical device that functions correctly but does not serve the patient. This reasoning leads to the first of the contributions proposed in the present thesis, which is the *"safety-first"* design principle presented in Section 3.2.1. This thesis aims to serve as a test-bed for this proposed approach, where exploring the airbag design process helps bridge the gap between current development efforts and certification.

2.3. Airbag impact attenuation: background knowledge

Airbag-type devices have a well-established, enduring research-development record with contributions found across multiple industries. The information presented in Sections 2.3.1 and 2.3.2 aims to represent the State of the Art practices adopted in airbag design. The scope of this investigation is developing a solid, although perhaps not comprehensive, understanding of design recommendations on which to base the proposal for the design workflow of later Section 3.2.2.

2.3.1. Established applications and common working principle

Airbag technology is well established across two primary sectors: *automotive* and *aerospace*.

In the automotive industry, these systems are implemented during vehicle crashes, distinguishing between general automobile crashes and motorcycle impacts. In the space industry, ever since the early space missions of the 1960s, airbags have been deployed during the final landing phases of space capsules. Within the aeronautical industry, specifically in the "specific" operations category, other niche applications have emerged, such as the experimental study of adaptive airbag systems for the protection of UAVs and general aviation vehicles [32] (following a long track record of exploration in the field of

adaptive technologies for impact absorption [33], [34]).

In either case, the primary objective of the airbag-type device remains the same: to be stored compactly in ordinary mission conditions, inflate within the allowable time before impact (whether during an emergency impact or landing), and decelerate the target body (whether a crash test dummy or a spacecraft) far more gently than in case of direct impact. When analysing these systems, purely analytical, hard-codable solutions are typically limited to single-degree-of-freedom (1-DoF) formulations, immediately resolving to the use of far more computationally-heavy finite element analyses to capture the complex two- and three-dimensional effects of the impact. The only exception to this trend is the technical report NATICK/TR-87/042 published by T. F. Patterson in 1985[2], which presents a 6-DoF analytical simulation built on closed-form component models, but remains inaccessible at the time of writing of the present literature search.

Despite the diversity of fields, all applications share identical fundamental working principles. An example of an analytical study of airbag-based impact events, and one of the most significant papers in shaping this thesis' work, is Zhou et al. (2019) [35], in turn heavily rooted in the SAND93-2133 technical report [36] on a code for predicting the performance of the Pathfinder lander's airbag impact attenuation system, which describes the core working principle of airbag physics in a one-degree-of-freedom impact event.

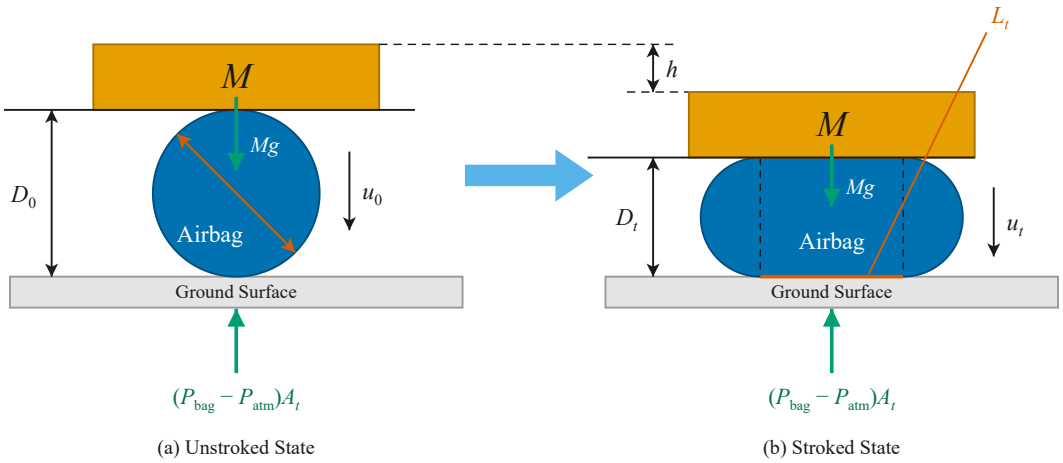


Figure 2.1: One-degree-of-freedom airbag impact model in the unstroked (a) and stroked (b) states, adapted from Zhou et al. [35].

The quantities under study for this type of phenomenon are shown in Fig.2.1, where a point load (yet, with a finite extension, as to avoid penetration in the airbag and justify the crude, though analytically necessary, *uniform pressure/control volume* assumption, where pressure is uniformly distributed within the airbag volume) is protected by a single inflated airbag in incipient contact with the ground surface after a free fall. The quantities are:

- M, the mass of the payload, which remains constant throughout the analysis and results in the vertically applied gravitational force Mg.
- u_0, the initial impact velocity at time t_0, which evolves to u_t after each subsequent time increment. It is updated at every time increment through kinematic analysis.
- D_0, the initial diameter of the airbag, which evolves to D_t following the same loop update concept. Broadly speaking, its meaning can be extended to the characteristic dimension of the airbag, according to the airbag shape selected for the test case among the available options (vertically- or horizontally-oriented cylinder, sphere, toroid, etc.).
- h, the *stroke*, i.e. the (vertical) distance covered by the payload when compressing the airbag during the impact event.

[2] T. F. Patterson, *Design, Fabrication and Testing of an Airdrop Platform Utilizing Airbags as Shock Absorbers*, NATICK/TR-87/042, U.S. Army Natick Research, Development and Engineering Center, Aug. 1985.

- $(P_{bag} - P_{atm}) \cdot A_t$, the normal reaction force from the ground, based on the difference between the internal pressure of the airbag P_{bag} and atmospheric pressure P_{atm}, multiplied by the footprint area A_t with the ground, calculated according to different geometric assumptions correlating $A_t = f(h, t)$.

The reference system is downward-positive, in accordance with the direction of gravity and the initial velocity, making the reaction force negative.

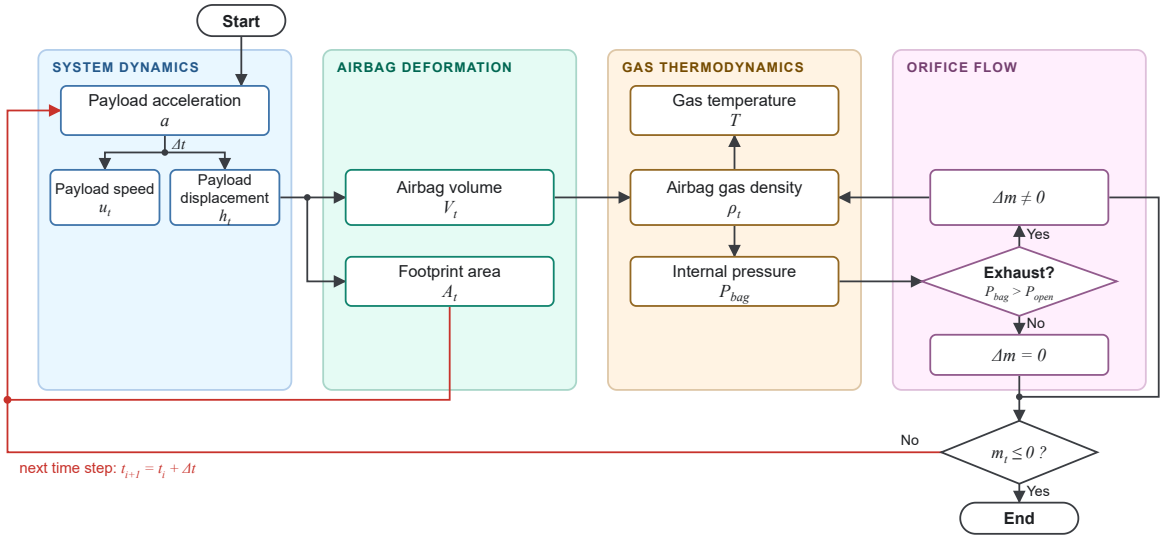


Figure 2.2: manca per caso la freccia di terminazione? Calculation flowchart of the one-degree-of-freedom airbag impact model, adapted from Zhou et al. [35].

Except for the initial conditions and the payload's mass, these quantities are updated iteratively at each timestep to produce characteristic outputs for airbag-governed impact events such as *airbag volume*, *internal pressure*, *internal gas mass*, *payload acceleration*, *payload speed*, and *payload displacement*. The logic of this calculation loop is schematized in Fig.2.2, which is an adaptation of the original diagram presented in the paper. The calculation loop is divided into four blocks, run sequentially at each timestep:

1. **System dynamics step**: The resultant acceleration a acting on the payload is calculated based on the reaction force from the ground calculated in the previous time step, and then integrated using the *central difference scheme NEEDS MORE DETAIL?* to obtain payload velocity u_t and displacement (or stroke) h.
2. **Airbag deformation step**: The airbag volume V_t and footprint area A_t on the ground surface are updated according to the kinematic information from the previous timestep. In this step, the hypothesis linking the stroke to the footprint area is enforced based on geometric assumptions, as mentioned earlier.
3. **Gas thermodynamics step**: The state variables of the gas are updated assuming adiabatic compression, as it is safe to assume that little to no heat is exchanged during the rapid compression process.
4. **Orifice flow step**: This critical step accounts for energy dissipation during the impact event. By modelling the outward airflow as flow through a one-way valve, treated as an orifice embedded in an impermeable fabric, typical nozzle fluid dynamics equations are applied to calculate the Δm vented at each timestep. In the case of this specific paper, mass is vented only if the internal pressure P_{bag} exceeds threshold pressure P_{open}, greater than atmospheric pressure P_{amb}. Within a single timestep, venting takes place as long as the exhaust condition withstands, updating gas density ρ_t and recalculating the gas' temperature T and pressure in an internal feedback loop.

In [35]'s framing, the code stops either when the manually-set duration of the phenomenon is reached, or when all mass is vented off. Testing of the code indicates that the no-rebound condition shown in the paper is numerically enforced by introducing an if-check managing the application of the reaction force.

The core work of the present thesis resides in turning the working principle of airbag impact events as presented in [35] into an informed, multi-variable problem that requires numerical optimisation. The code presented in later Section 4.5 is an expanded version of this implementation, incorporating additional criteria and alternative termination mechanisms.

2.3.2. Design guidelines, requirements, and variables

Framing

In essence, an airbag can be considered as an energy absorber and dissipator that must bring the payload from its impact velocity v to rest within an available stroke h, without exceeding an allowable deceleration a. This relation is expressed in its most basic form by the kinematic relation [37], [38]

$$h = \frac{v^2}{2a} \tag{2.1}$$

which shows the central trade-off in airbag design: the lower the allowable deceleration, the longer the stroke needed to decelerate the payload and, therefore, the larger the airbag. During compression, the airbag behaves as a non-linear spring whose stiffness depends on its geometry and venting characteristics [35], [36], [39]. Venting is what dissipates the absorbed energy [39]; whatever energy remains stored in the gas at the end of the stroke is returned to the payload as rebound [40]. Increasing the volume of the airbag and inflation pressure raises its energy dissipation capacity, but at the cost of higher membrane loads on the fabric [37], [39].

Configuration concepts

Airbags are generally classified into three types depending on whether they contemplate the presence of venting in their design: *non-vented*, *vented*, and *combined* [35]. A combined airbag usually consists of an outer, main-venting airbag and an inner, non-venting, anti-bottoming airbag. This "bag within a bag" concept prevents the payload from bottoming out once the outer bag has vented [39], [41], [42], and commonly relies on complex sequential inflation of the outer and inner bags [42]. An alternative to this design is using multiple airbags side by side rather than one within the other, following a "fail-safe" design philosophy by sharing impact loads and providing redundancy in case of airbag failure [37]. Rather than completely enveloping the payload, the airbag can act as a buffer cushion beneath it [42], or inflate between plates attached to the sides of the cargo [43]. Moreover, venting can be passive, activated pneumatically, or active, activated electronically [34], [40]. Active venting is the basis of adaptive impact absorption, in which the airbag responds to the actual, case-specific impact load [32]: for example, in multi-chamber designs, controllable valves allow gas to flow between adjacent chambers and to the environment, redistributing a non-uniform load across the structure [33], [34].

Design requirements

Notable airbag design requirements and recommendations identified in literature are summarised in Table 2.2 and relate to impact conditions, such as the vertical and horizontal velocity, impact attitude, and landing surface [37], [39], [44]; to the allowable deceleration of the payload; and to general constraints on mass, packaging volume and deployment time [32], [42], [45].

Table 2.2: Design requirements for airbag landing and impact-attenuation systems identified in the literature.

Category	Requirement	Sources
Dynamics	Limit the peak acceleration transmitted to the payload	[32], [39], [42], [46], [47]
	Reduce the touchdown velocity to a target value	[32]
	Cope with horizontal velocity (non-nominal impact attitudes coupling)	[37], [39], [44]
Landing conditions	Cope with different landing surfaces	[41], [44], [48]
	Avoid fabric rupture and tearing	[39], [45]
	Prevent bottoming-out of the payload	[35], [39], [41], [42]
	Limit rebound and prevent rollover	[40], [42]

continued on next page

Category	Requirement	Sources
Inflation pre-landing	Synchronise venting and payload contact	[49]
	Deploy fully within the available time	[32], [42]
	Retain pressure for the required duration	[42], [45]
Packaging	Survive long-term stowage in the packed state	[45]
	Integrate with the host vehicle without interference	[32]
	Withstand the environmental temperature range	[44], [45]
General	Minimise system mass	[32], [42], [45]
	Allow reuse	[42], [44]
	Provide redundancy against single-bag failure	[37]

Design variables

All airbag design variables found in the literature analysed for the present review can be found in Table 2.3. The variables in the table are divided into two umbrella categories: *Energy absorption at impact* and *Deployment and packaging*. These categories can be mapped almost directly onto two distinct impact phases (at-impact and pre-impact behaviour, respectively) and, consequently, into two separate, although not entirely uncoupled, design stages: *phase 1*, focusing on the inflated configuration during impact to *limit peak acceleration* below the allowable limit set by the payload's integrity requirement, and *Phase 2*, focusing on the pre-impact inflation process by mostly *optimising out-of-stowage deployment time* and leading to prototyping.

Table 2.3: Airbag design variables identified in the literature, grouped by design focus and subcategory, with the associated trade-offs.

Group	Variable	Trade-off	Sources
Energy absorption at impact			
Geometry	Shape	Pressure-vessel shapes carry membrane loads well; faceted shapes are easier to build	[35], [37], [39], [41], [48], [50]
	Characteristic size	More volume absorbs more energy but raises membrane loads, chance of rebound, and overall structural weight	[37], [39], [40], [51]
	Payload contact area	Larger contact spreads the load; wrapping the entire payload wastes material and increases weight	[41], [42], [50]
	Number of chambers	Inner anti-bottoming and multiple bags add margin and redundancy (fail-safe philosophy), but increase weight and complexity	[34], [35], [37], [39], [41], [42]
	Tethers and webbing	Divert inertial loads from the fabric and control the shape, at the cost of added parts	[41], [48], [52]
Fabric	Fabric type	Strength, tear resistance, low permeability, temperature flexibility vs. mass, packability	[41], [42], [45], [47], [48], [49], [52], [53], [54], [55], [56]
	Fabric thickness	Thicker fabric raises the rupture margin but slows deployment, enlarges the package, and increases weight	[39], [46], [57]
Venting	Vent area	Larger orifices dissipate more energy and limit rebound, but risk bottoming-out	[36], [37], [39]
	Vent location	Sets the outflow and whether the ground or payload blocks the vent during the stroke	[41]
	Vent burst pressure	Early opening lowers the peak pressure and resultant acceleration but spends stroke; late opening raises the peak pressure and acceleration	[37], [49]
	Fabric porosity	Distributed gas leakage dissipates energy without vents but reduces pressure retention	[45], [46], [57], [58], [59]
	Vent actuation	Active valves adapt to the actual impact; passive valves need no sensing or power	[32], [33], [34], [40], [41], [42]
Gas	Initial bag pressure	Higher pressure stops the payload in a shorter stroke but raises membrane loads	[37], [39], [49]

continued on next page

Group	Variable	Trade-off	Sources
	Gas composition	Specific-heat ratio sets compression stiffness; mixtures cut temperature sensitivity	[41], [47], [49], [53], [60]
Deployment and packaging			
Inflation	Inflator type	Pyrotechnic vs. stored compressed gas: drives gas output, package size and triggering mechanism	[41], [49], [53], [60], [61]
	Inflator supply pressure	Faster filling vs. higher deployment loads; the flow chokes at sonic conditions	[49], [58]
	Inflator mass flow rate	Shorter injection time vs. higher deployment loads	[46], [57], [58]
	Inflator gas temperature	More pressure per unit gas mass vs. thermal load on the fabric just after inflation	[41], [45], [46], [57]
Housing	Housing size and geometry	Compact housings save mass and volume but pack the bag tighter	[46], [62]
	Housing material	Plastic saves mass over metal; metal bases give a stiff mounting	[38], [46], [53], [62]
Folding	Folding pattern	Governs unfolding kinematics, deployment time and pre-stress in the packed fabric	[45], [55], [57], [58], [62], [63], [64], [65], [66]
	Number of folds	Smaller package vs. longer, less predictable unfolding	RIVEDI LIBRI DI MATEMATICA DELLA CERAGIOLI
Seams	Seam locations	Seams concentrate stress and are the main failure site; patterning minimises their number	[45], [52], [67]

The table conveys the complexity of the design problem at hand, highlighting the large number of design variables. Multi-variable problems typically require an optimisation process where an optimal combination of variable values is achieved through guided iterations based on penalty concepts. This consideration forms the basis for the following chapters, in which the airbag design process is distilled into an ad-hoc workflow tailored for a specific payload application and impact scenario, focusing on a single design phase among the two presented. It is now possible to define the research-development objectives of the present thesis, leading to the proposed solution in Chapter 3 and the specific implementation choices presented in Chapter 4.

2.4. Research-Development Objective

Having presented the context in which the research project is allocated and having gathered the necessary design information, it is possible to address the requirements set by the project commissioners:

- Identify and characterise the mechanical allowables of typical medical payloads based on general packaging requirements for UAV-transported medical goods;
- Develop an informed conceptual design of an airbag system capable of deploying during emergency release conditions to absorb impact energy and preserve payload integrity.

These requirements can be summarised in the following Research-Development Objective and Research Questions:

Research-Development Objective

Develop and validate a design workflow to model and simulate an airbag-based protective system for UAV medical payloads using LS-DYNA, with the primary objective to ensure the structural integrity of pharmaceutical materials and their containment during drone failure and emergency payload release, thereby preventing environmental dispersion and supporting certification requirements.

Research Question(s)

How can a systematic multi-phase simulation workflow be developed to design an airbag-based protective system for medical payloads during UAV emergency-release events?

1. What are the allowable mechanical limits (acceleration, stress, etc.) for medical products during emergency release, and how should these define the design envelope?
2. How can the inflated airbag configuration be optimised to satisfy impact-absorption requirements while balancing competing design requirements?
3. How can the optimised inflated geometry be transformed into a compact and functionally deployable system?

3

Methodology: the Airbag Design Workflow

3.1. Problem statement

To properly outline and, more importantly, test the workflow proposed as a main objective of this thesis, it is necessary to identify the specific context in which this project is embedded, as it originates from prior research work conducted within the specific framework of Turin. The design inputs presented below were gathered from analysing the results presented by the group ([9], [10], [11], [12], and [13]) and categorising them into mission aspects on which to root the proposal of Section 3.2.

3.1.1. Reference mission and design inputs

Route and flight profile

The reference mission is based on a hospital-to-hospital route as initially suggested by Cavallo (2020) [9], connecting two hospitals across the city of Turin as shown in Fig. 3.1, representing a city map with route schematics. In particular, the starting point is the Molinette Hospital in the southern region of the city, and the destination is the San Giovanni Bosco hospital in the northern sector. As they are located far apart, these two hospitals require medical deliveries to traverse a broad part of the city's territory. The hospitals are connected through two possible routes:

- **Route A**: Initially hypothesised by Cavallo and confirmed by Bertelli (2021) [10], it takes advantage of the above-water air corridor over the river Po and its tributary Stura, mostly skirting the city and flying over city ground only in the relatively short segments connecting the hospitals to the rivers. Among the two routes, it is the safest (and, thus, most certifiable according to Cavallo) as it reduces the probability of ground impact in case of drone failure due to the UAV's permanence in the above-water air corridor. This route was selected as viable within the Concept of Operations (ConOps) that informed the preliminary drafting of the SORA. Approximate length: 10 km. Approximate duration time: 15 min, comparable to ambulance-based delivery times.
- **Route B**: A crude alternative to Cavallo and Bertelli's Route A, connecting the two hospitals in a straight line over the city. Angelini (2023) [13] deemed this route infeasible based on their comparative safety analysis of delivery means, primarily due to the lack of proper risk mitigation measures concerning in-flight failures. However, as the approach proposed in this thesis aims at partially mitigating these risks, it was chosen to consider this route (at least for preliminary evaluation).
- **Alternative routes *(omitted from analysis)***: Fiorentino (2021) [12] and Angelini (2024)'s [13] work is rooted in minimum risk-based route selection considering territories that extend beyond city borders, specifically including the entire province[1] in case of Angelini's study. Either using the

[1] In Italy, local government is organised across three administrative levels: regions, provinces, and municipalities. Here, assessing the Province of Turin corresponds to evaluating the territorial sub-region of Piedmont administered by the provincial

RRT* algorithm in Fiorentino's case or the Dijkstra and A* algorithms in Angelini's case, these two studies share a common approach of considering multiple layers (such as population density, shelters, and critical infrastructure CHECK) to calculate the probabilistic minimum-risk paths for different means of transportation (eVTOL, UAV, etc.), without predefined start and end points, thereby determining the optimal drone hub location across the province. valid for FIORENTINO TOO??? The results of these works were omitted from the definition of the reference mission for two main reasons. First, they do not specifically target hospital-to-hospital operations; their scope focuses on optimising general logistics rather than facilitating inter-hospital transport, contrarily to one of the primary objectives of the present thesis. Second, their findings indicate that drone operations, when evaluated against the merit indices adopted in their analyses, are deemed unviable due to safety constraints. In contrast, the objective of this research is precisely to identify a mission scenario that is unfeasible under current conditions and implement risk-mitigation strategies to ultimately enable drone transit.

In either Route A or B, as specified in Bertelli's ConOps, the cruising height is about 75 m, at 17 m/s cruising velocity (approximately 70% of the maximum cruising speed allowed by the UAV platform of 25 m/s, unsustainable for the entire duration of the mission at the time of evaluation), with possible speed variations according to mission urgency if specified in the ConOps.

Accounting for the presence of wind in the operating environment, wind speed in Turin averages about 1.8 m/s (6.5 km/h) [68], with a peak of about 28.6 m/s (103 km/h) in 2013 [69].

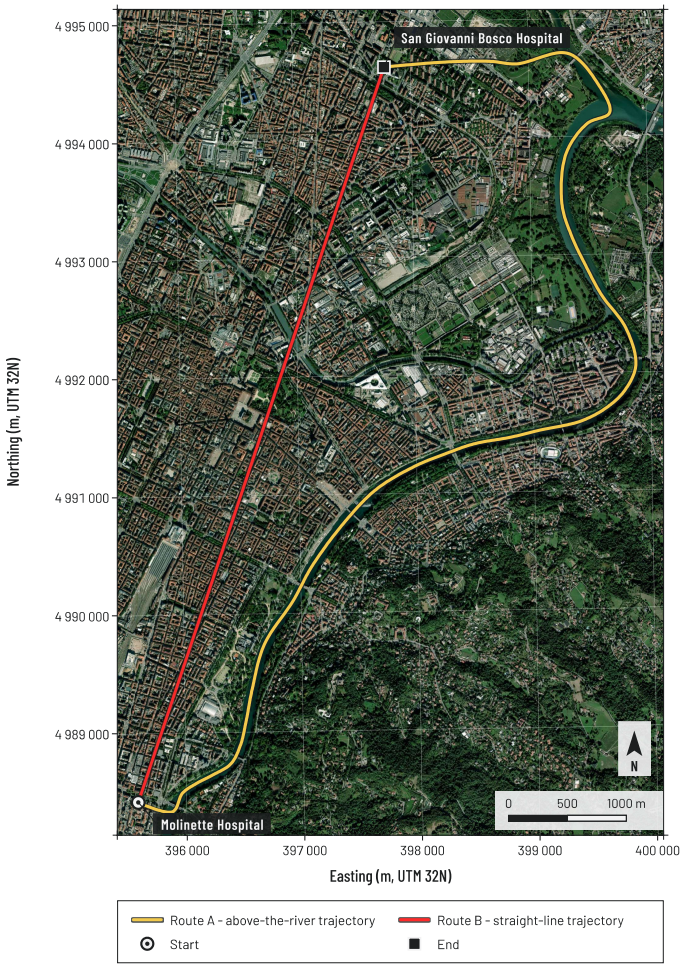


Figure 3.1: Reference mission routes across Turin, from the Molinette Hospital to the San Giovanni Bosco Hospital: Route A (above-the-river trajectory) and Route B (straight-line trajectory).[2]

capital of Turin.

UAV Platform

The UAV platform choice is inherently linked to the reference mission specifications above, meaning that the platform is required to sustain the cruising speed requirements specified in the ConOps. A Maximum Takeoff Mass (MTOM) of 9 kg and a 0.81 m characteristic dimension were also specified in Bertelli's ConOps, leading to the suggestion of a potential drone model suitable for the situation: DJI's *Matrice 300 RTK*. Despite having characteristics suitable for the speed requirements, the drone has a payload limit of 2.7 kg and a wind resistance of 15 m/s in cruise. While this drone was initially selected for the preliminary SORA, its specific parameters are not binding for the present study. In particular, the payload-mass and wind-resistance limitations may be disregarded, given that the ConOps set a much broader requirement for MTOM. In addition to these requirements, Cavallo, in their UAV platform market evaluation, considered several other factors: maximum flight time (usually corresponding to half the autonomy), geometry, glide speed, and frontal area, the latter covering an important role in free-fall drag calculations. By ConOps, the platform is equipped with a parachute, whose contribution is ignored in the calculations regarding the free-fall event.

Failure scenario and impact conditions

The chosen failure scenario implies an in-flight malfunction or fatal failure of the UAV in combination with a failure of the parachute mounted on the drone by ConOps. Excluding the nominal, non-conservative parachute descent of 3 m/s estimated by Bertelli, in this extreme failure scenario the payload is dropped in free air, separated from the drone at an altitude of 75 m above ground. After the failure, a *ballistic* descent mode is chosen for calculations of impact time and point depending on the release position on the map, including the effects of gravity and aerodynamic drag (as opposed to *uncontrolled glide* descent mode, which is applicable to the UAV platform but not to its detached payload), used in Angelini's impact probability study for ground risk evaluation and inspired by PRIMATESTA e i suoi predecessori CHECK!!!. Wind enters the calculations (as mentioned in the reference route definition) as a horizontal velocity component.

Payload container

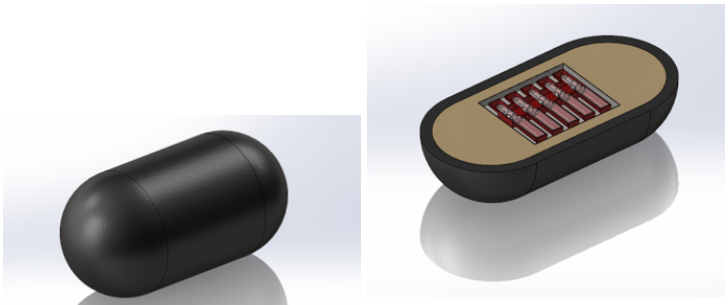
The design of the payload container to be carried by the UAV platform in delivery missions was proposed, but never fully prototyped, by Sivieri (2021) [11]. Its geometry is shown in Fig. 3.2 as modelled for the scope of the present thesis: the central body is a cylinder capped with two hemispherical domes. In Sivieri's original design, this container had to be constructed via stratifying four layers, from inside out: first layer of 3d-printed PET-G with thickness 2 mm, second layer of manually applied thermal insulation of unknown effective thickness, third layer of 3D-printed PET-G and 2 mm thickness, and a final layer of manually-applied thermo-reflective paint. This sandwich-type structure was designed specifically for evaluating the thermal and vibration loads of a reference mission similar to the one presented earlier, and thus avoiding making considerations on transportability, attachment mechanisms, and other application-specific parameters. From a structural perspective, the first and third layer of Sivieri's design were connected by four radial ribs, but this detail was deliberately omitted in the exemplification of the payload container for modelling simplicity and redundancy, as explained later in Section 4.4.

Due to the artisanal nature of this design, linked to the difficulty in modelling specific 3D-printed PETG and the unquantifiable specifications of the insulation material, the payload container was modelled by retaining its shape with a 5 mm thick shell [verify thickness in the FE model] of injection-moulded Acrylonitrile butadiene styrene (ABS), commonly used in packaging. The cavity is filled with polyurethane foam, chosen to simulate a protective wrapping (such as bubble wrap) that medical operators would use as established praxis to mitigate vibrations. The thermal paint is omitted from the payload container modelling process.

Payload

Considering the wide variety of medical goods potentially suitable for UAV-based air transport, it was necessary to identify a payload representative of standard shipments, excluding human organs and specialized equipment (e.g. defibrillators) to focus specifically on pharmaceuticals. Therefore, glass

[2]Imagery: Esri, Maxar, Earthstar Geographics, and the GIS User Community. Basemap: Web Mercator; grid: WGS84 / UTM 32N.

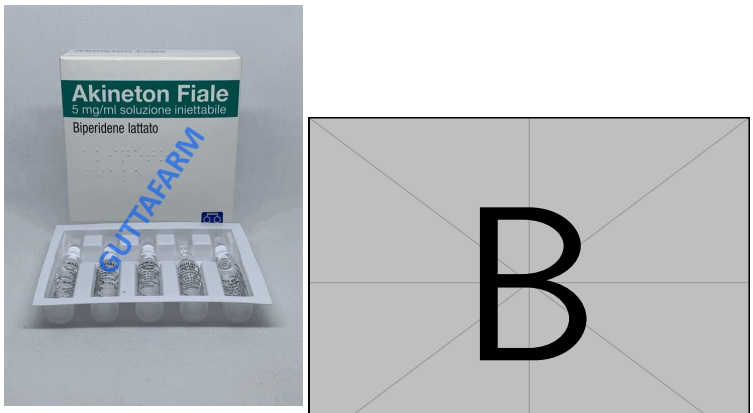


(a) Full view of the closed container.

(b) Sectioned view showing the outer shell, the foam filling, and the housed payload.

Figure 3.2: IT's REALLY UGLYPayload container geometry: cylindrical central body capped with two hemispherical domes.

phials of the medicinal product *Akineton Fiale*, containing *biperidene lactate* at a concentration of 5 mg/ml in injectable solution and shown in Fig.3.3[3], were chosen. The product is commercially supplied in boxes of five phials, each individually packaged in a plastic blister tray.



(a) Akineton Fiale, 5 mg/ml injectable solution, as supplied in its original packaging.

(b) Single Akineton phial.

Figure 3.3: Akineton Fiale packaging and single phial.

Akineton is a pharmaceutical product commonly used in the treatment of Parkinson's disease and drug-induced extrapyramidal symptoms, and is usually administered by intravenous or intramuscular injection. For the purposes of the present thesis, only the mechanical properties of the glass container are of interest, as the pharmacological action of the medical fluid is not relevant to the type of structural analyses implemented in this study.

3.1.2. Information gap

The only missing inputs required for a full design workflow evaluation are:

- The derived *impact velocity*, angle, and mass for the selected scenario;
- The quantification of the phials' fragility through the characterisation of their material's mechanical *allowables*;

These failure scenario data must be fed into the design loop(s) proposed below to enable an effective design of the airbag-based impact attenuation system.

[3]Source of photo: `https://www.guttafarm.net/akineton-cena`.

3.2. Proposed approach

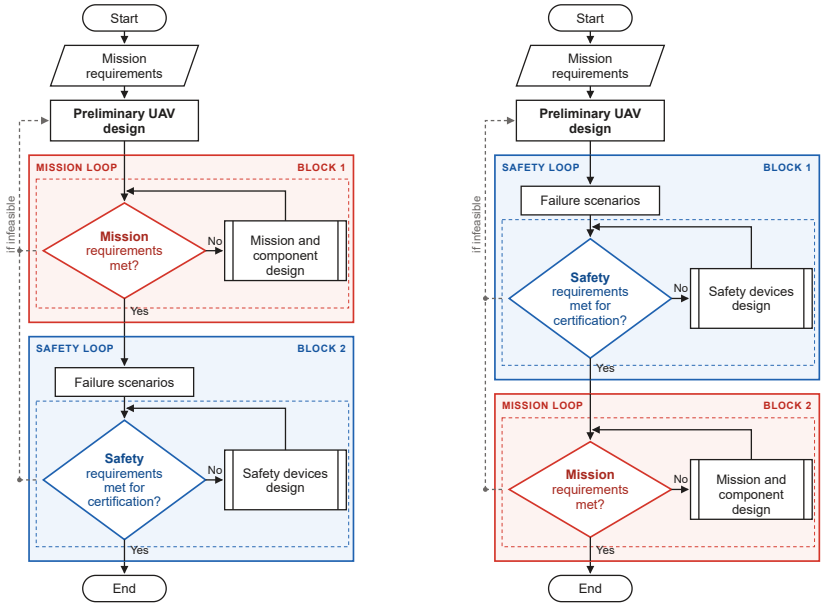
3.2.1. Outer design loop: formalising the Safety-First principle

In response to the gap identified in Section 2.2, the first cotribution of this thesis is to propose an alternative design approach adopting (and later testing) the following guidelines:

Safety-First Design Principle

1. The airframe and payload must be considered as two distinct entities;
2. In the type of operations considered, saving the payload takes higher priority than saving the airframe (carrying organs, expensive vaccines, emergency medicines, or high-value medical equipment is worth more than the drone itself);
3. Overall, the objectives are to minimise or mitigate:
   - ground impact consequences,
   - payload damage (preserving its integrity),
   - environmental dispersion;
4. These considerations should be applied at the beginning of the design process, possibly by integrating multiple safety devices (e.g., an airbag and a parachute working in tandem) and adjusting flight-mechanics considerations accordingly.

A direct consequence of applying the Safety-First design principle in UAV-based operations' design is changing the sequence of design events, as exemplified in the simple flowcharts of Fig. 3.4, where Fig. 3.4a and Fig. 3.4b compare scenarios in which the mission loop can be interchanged with the safety loop to achieve design convergence.



(a) *Mission-First* approach **(b)** *Safety-First* approach

Figure 3.4: Sequence of design events in the two approaches: *mission-first* and *safety-first* loops.

3.2.2. Inner design loop: a payload-protection subsystem design workflow

This section aims to expand the *Safety loop* concept presented in Section 3.2.1 by informing it with the design guidelines distilled in Section 2.3.2. The proposed workflow can be seen in Fig. 3.5, and is divided into seven steps that bridge the gap between payload allowables' evaluation and safety device

(airbag) prototyping. The loop is also split into two main branches, highlighting the boundary between the present MSc Thesis' deliverable and steps left to future developers. The first branch (steps 0-4) is partially payload-specific, meaning that some of the considerations made throughout these steps depend on the specific medical payload considered, while also generalisable to a broader spectrum of medical payload receptacles; the second branch (steps 5-7) are exclusively geometry-specific, implying that they require information on receptacle design, UAV-container hooking mechanism, and general information on system bulk, making them unattainable for the DIMEAS feasibility studies group, as they currently lack access to a functional medical payload container beyond their initial artisanal prototype by [11]. Completing the workflow should lead directly to the Mission loop as shown earlier in Section 3.2.1.

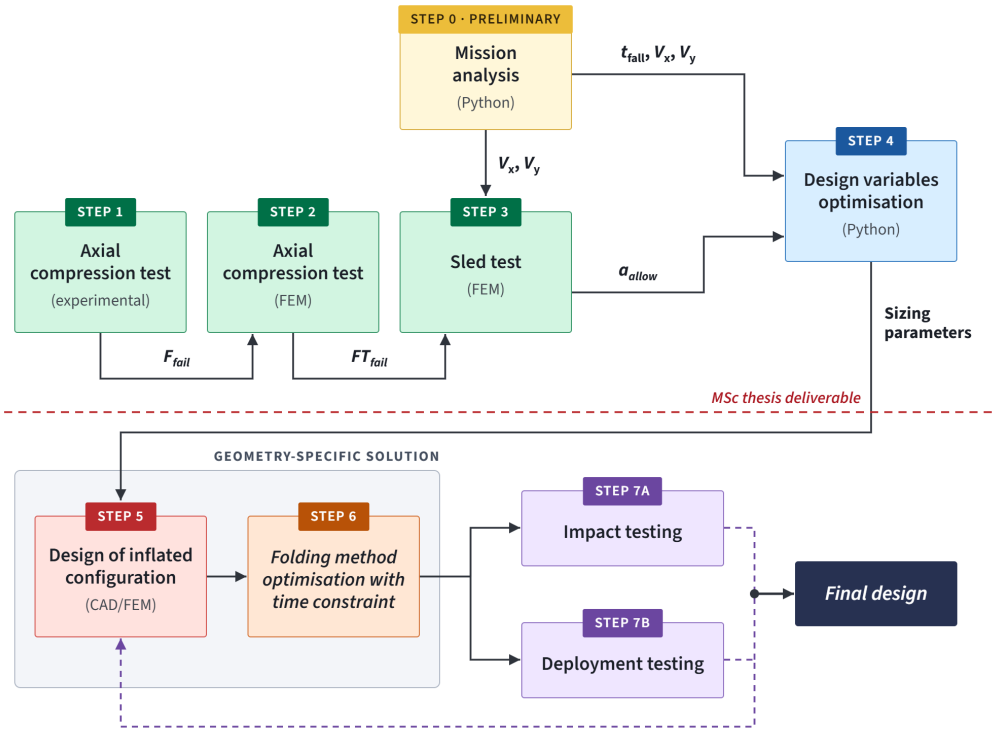


Figure 3.5: Inner design loop, specific for the airbag-based payload-protection subsystem.

Step 0 - Preliminary mission analysis

This step covers the preliminary analysis of the payload's impact conditions. It evaluates the drop scenario, incorporates statistical analysis of wind direction and cruising conditions, and computes the trajectory and landing location of the payload. Serving as a general free-fall analysis, this phase yields key design inputs: the free-fall time (t_{fall}) and the horizontal and vertical impact velocities (V_x, V_y respectively).

Step 1 - Axial compression test (experimental)

This step covers the experimental characterisation of the glass phials' (the selected medical payload) breaking point through laboratory testing. The goal is to conduct a test campaign to characterise the phial's mechanical behaviour, specifically providing key data, such as the failure force (F_{fail}, better defined in Section 4.2), which feeds into the next phase to accurately calibrate the Finite Element Method (FEM) model.

Step 2 - Axial compression test (FEM)

This step focuses on matching the experimental behaviour observed in Step 1 with a Finite Element Method (FEM) model to accurately reproduce the material's mechanical response. Specifically, both experimentally and in the FEM model, the phials are subjected to axial compression loads, resembling the stress type to which they would be subjected to during an impact event. To replicate the laboratory

results and build a calibrated material card that captures the phial's failure behaviour, the simulation mirrors the physical testing setup as closely as possible. This yields a generally realistic phial model, complete with its brittle failure threshold (FT_{fail}), that can be integrated into the full sled-test impact simulation in Step 3.

Step 3 - Sled test (FEM)
This step involves simulating a sled test, heavily inspired by automotive industry practices, to observe the mechanical response of a test article (such as a dummy in vehicle crash testing, or in this case, a glass phial) measured through perceived accelerations and damage under an input acceleration profile. The simulated test subjects the payload container (based on the artisanal design proposed by [11]) to realistic impact scenarios by reproducing an acceleration profile equivalent to a free-fall impact under the conditions defined in Step 0. This approach relies on the principle that the accelerations experienced by the payload directly correlate with the kinetic energy that must be dissipated during impact, linking the input acceleration profile to the resulting output. The primary output of this phase is the payload's allowable acceleration (a_{allow}).

Step 4 - Design variables optimisation
This fourth and final step of the thesis deliverable translates the established boundary conditions evaluated in previous steps (failure loads, allowable peak accelerations, impact velocities) into a numerical Python script for airbag sizing. Building upon the framework established in Section 2.1, the simulation optimises the airbag design against critical criteria, including total mass, bottoming-out speed, and rebound behaviour. Ultimately, this tool yields fundamental sizing parameters that will inform the future geometry-specific design of the system.

Steps 5-7: Design of inflated configuration, folding method optimisation, impact and deployment testing
Steps 5 and 6 shift the focus to geometry-specific development. Step 5 details the fully inflated configuration tailored directly to the geometry of the payload and UAV, matching the gas generator mechanism to the drone layout and impact timelines. Step 6 then builds upon this complete configuration by incorporating real-world manufacturing constraints, such as folding patterns, seaming details, and stowage volume, to optimise the deployment mechanism. While inherently interconnected, Step 6 specifically targets deployment time optimisation, as defined in Table 2.3.

Once a preliminary design and folding mechanism are established, the prototype undergoes a two-part experimental validation campaign: Step 7A (deployment testing) and Step 7B (impact testing). This phase validates the FEM models of steps 5 and 6 by subjecting physical prototypes to real-world impact conditions, operating in an iterative feedback loop between simulation and testing until design convergence is achieved.

4

Results: through the Workflow

This chapter presents an example of how the airbag design workflow can be applied following the steps presented in Section 3.2.2. Each step requires its own ad-hoc procedure and analysis of the results. Hence, the procedure of each step is described first, its results are presented, and these are then fed into the next step. This structure is meant to show how the output of each step is used in the next, starting from a preliminary mission analysis, continuing with the characterisation of the medical payload and resulting allowables, and ending with recommendations on the design of an airbag intended to protect the payload in case of a drone failure.

The inputs used in this example are partly experimental and partly hypothesised. The material model of the glass phials is calibrated against experimental compression tests, while the geometry of the payload bay and the behaviour of the sponge are representative assumptions. Therefore, the numerical results should be read as a demonstration of the workflow, not as the qualification of a specific design.

4.1. Step 0 - Preliminary mission analysis

4.1.1. Implementation

Map extraction and preprocessing

First, it is necessary to conduct a territorial analysis of the area above which the mission is supposed to take place. This implies processing OpenStreetMap (OSM) data extracted for the relevant geographical area of Turin. The OSM file is usually a plain XML document containing three types of elements, namely nodes, ways and relations. Nodes are individual points with a latitude and a longitude, ways are lines or polygons connecting the nodes, and relations are groups of ways that together form larger features, like a river.

A code was necessary to extract this information and classify it, so that the territory of Turin could be divided into five different categories.

1. Water, including waterways such as the Po river or the Dora Riparia, which are described as ways with a thickness, going from 60 m for the Po to 18 m and 9 m for smaller waterways [?].
2. Buildings, i.e. anything tagged with the `building` key in the OSM file, for example schools, churches, houses, hospitals, etc.
3. Roads, i.e. the ways tagged as `highway` across the city. Any corridor [?] or railway line is also classified as such.
4. Green/natural. The idea here was to classify all natural terrain, such as gardens, pitches and parks, but also grass, farmland, meadows, recreation grounds and woods, i.e. anything that has to do with nature.
5. Ground, which is where everything else that is not in the first four categories falls. It is assigned to any cell that is still unclassified at the end of this block.

In this way, all the individual features of the map are classified into these five categories.

After this initial classification, the map is built with a first conversion of the reference system, because in the OSM file the coordinates are in longitude and latitude, while we need them in meters. A UTM projection (zone 32N [EPSG:32632?]) is used to convert from one to the other. In particular, it makes the angular geometry planar, so that what was a globe becomes a plane, in a way that depends on the area of the globe in which it is. Since Turin is in zone 32, the coordinates are projected onto the plane of that zone and are now in meters, about 4990 km north of the equator and about 100 km west of the central meridian of the zone (9°E). The bounding box measures approximately 4757 m (easting) × 6924 m (northing). At the current 500 × 500 resolution (250 000 cells), each cell measures approximately 9.5 m × 13.8 m.

Once the OSM data is classified and everything is converted into UTM meters, the third step is to actually build the grid. A 500 × 500 grid is built, and each cell is assigned one of the five categories (ground, building, etc.) depending on what lies at its centre. If the centre falls on a tree, the cell is assigned to green/natural, if it falls on a building, to buildings, and so on. The idea is to simplify the process of building the actual grid, without considering the variation within the single cell.

Step 4 is then to visualise the grid as a map that can actually be analysed, as seen in Fig. whatever.

Mission trajectory definition

The second step is to define the trajectory above the map produced earlier. The idea is that the waypoints of the trajectory, selected manually and set up in latitude and longitude in an external Excel file, are loaded and checked against the bounding box of the map, and any waypoint falling outside it is discarded. The waypoints are then converted to UTM meters (zone 32N, EPSG:32632), with the same transformation used for the map, and interpolated with a cubic B-spline using the `splprep` function of SciPy. No smoothing is applied ($s = 0$), so the curve passes exactly through every waypoint. The spline is parametric, i.e. x and y are both expressed as functions of a parameter $t \in [0, 1]$, and the curve is made of cubic polynomials joined at the waypoints with continuous first and second derivatives. If fewer than four valid waypoints are available, the code falls back to straight line segments ($k = 1$).

Then, points every 10 m along the curve are extracted through an arclength parametrisation. The spline is evaluated at 100 000 equally spaced values of t, and the cumulative arclength is computed by summing the lengths of the straight segments between consecutive points. The values of t that correspond to every 10 m of distance are found by linear interpolation on this cumulative arclength and saved, and the spline is evaluated at these values to give the points where the evaluation is conducted later. The same goes for the heading angle, computed at every sample point from the local direction of the trajectory and measured counterclockwise from East. The sample points are accurate because, for a total length of about 11 km, consecutive points of the dense evaluation are about 0.11 m apart, so there are about 90 of them for every sample point.

Finally, the spline is visualised on the map and the trajectory points are saved.

Failure and fall model

The full model assumes that the payload is detached from the drone and then falls without propulsion or control, so the equations of motion that determine its trajectory include both the gravitational contribution and the aerodynamic contribution, the latter through a drag term. Given the air density ρ, the frontal area of the payload S and the drag coefficient C_d, the drag force is

$$F_d = c\,V^2, \qquad c = \frac{1}{2}\,\rho\,S\,C_d, \tag{4.1}$$

where V is the instantaneous speed of the payload.

For every combination of parameters, the drag constant c is calculated first, and then the terminal velocity

$$\Gamma = \sqrt{\frac{m\,g}{c}}, \tag{4.2}$$

i.e. the maximum vertical speed the payload can reach, at which drag balances gravity. For the payload considered here ($m = 1.5$ kg, $S = 0.04$ m^2, $C_d \approx 1.0$), $\Gamma \approx 24.5$ m/s, which is close to the cruise speed of the drone (about 23 m/s).

The descent is then divided into three phases. There is an upward phase, present only if the payload has an upward vertical velocity at separation. Then there is a descent in which the vertical velocity is lower than the horizontal velocity, which ends when the two become equal. Finally, there is the final descent from this crossing to the ground, in which the vertical velocity catches up and approaches Γ while the horizontal velocity keeps decreasing. Each phase has its own approximate closed-form expressions for position and velocity **[key?]**, and the horizontal range is the sum of the distances covered in the three phases. The effect of the wind and of the drag is taken into account throughout [how does the wind enter the equations?].

The conditions at the moment of separation are not known exactly. For this reason the uncertain parameters are modelled as Gaussian distributions, each described by a mean μ and a standard deviation σ, and sampled randomly at each simulation, as listed in Table 4.1.

Table 4.1: Uncertain parameters of the drop model, each sampled from a Gaussian distribution.

Parameter	Unit	μ	σ	Remarks
Altitude at separation h	m	[?]	[?]	Small variations in flight altitude
Drag coefficient C_d	-	[?]	[?]	Payload tumbling or rotation; one value shared by all the samples of the same call [?]
Horizontal velocity at separation $V_{x,i}$	m/s	[?]	[?]	Inherited from the drone cruise speed
Vertical velocity at separation $V_{y,i}$	m/s	0	[?]	Zero mean assumes level flight at failure
Wind speed V_w	m/s	[?]	[?]	
Wind direction ψ_w	deg	[?]	[?]	A large σ produces a wide lateral spread

Monte Carlo impact loop

For each point of the trajectory calculated earlier, i.e. for every value t^* corresponding to a distance of 10 m along the curve, a Monte Carlo simulation is run with 5000 combinations of parameters, selected randomly from the Gaussian distributions of Table 4.1. The idea is that, once these parameters are chosen and the heading at that point is read from the trajectory, the function `impact_point_ballistic_descent` is called, which calculates where each of these failure scenarios ends up, i.e. where the payload falls in case of that scenario. The function computes the horizontal range and the impact time of all 5000 scenarios at once and rotates the resulting cloud of impact points by the heading, so that it is oriented along the flight direction. Each impact point is then shifted by the wind drift $V_w\, t_{im}$ in the wind direction, assuming that the wind carries the payload at its own speed for the whole duration of the fall, and the cloud is finally translated to the UTM coordinates of the trajectory point.

The impact points are then counted on a grid of cells [$N_1 \times N_2$?] that are much smaller than the cells chosen to represent the terrain map, so the impact distribution is computed with a much higher accuracy. For each trajectory point this gives a local distribution, and the local distributions of all the points are summed up with equal weight. The sum is normalised over the total number of impacts that have happened, i.e. 5000 times the number of points on the trajectory. Each cell then contains the probability that, in case of a failure along the trajectory, the payload falls in that cell.

Finally, each impact cell is associated with the terrain category of the nearest cell of the terrain map, and the probabilities of all the cells belonging to the same category are summed up. In this way the total probability of impact on the ground, on water, on buildings and so on is obtained.

4.1.2. Results

Convergence of the Monte Carlo
taking statistically significant number of samples, and also considering a statistically significant number of points above or below high or low percentiles (e.g. if there's only 1 data point above the 99th percentile, then you don't consider it)

Impact location distribution
Impact conditions
Impact surface statistics
Design inputs for the following steps
Limitations
Step 0 deliverable: impact velocities and free-fall time

4.2. Step 1 - Axial compression test (experimental)

4.2.1. Implementation

The pharmaceutical product selected for the illustrative characterisation of medical payloads for UAV-based transportation is one of the most fragile phial types currently available on the pharmaceutical market. This choice was deliberate: by characterising a worst-case, low-strength test article, the resulting airbag design is expected to provide adequate protection also for more robust payload cases. The test consisted of a simple compression test, through which the maximum compressive load sustainable by the phial was measured.

Therefore, the parameter of interest for this test was the compressive force at which the phial broke, hereafter referred to as the ***failure load*** F_{fail}, expressed in Newtons. This was the only quantity of interest measured during the test. The failure load constitutes the experimental datum that is used to calibrate the finite element model (FEM) of the compression test in the design workflow's Step 2.

Test Article
The test articles were glass phials containing 5 ml of saline solution, 70 mm in height, 15 mm in diameter, with an assumed wall thickness of 0.4 to 0.5 mm (see Fig.4.1) as shown in Fig. 4.1. The saline solution phials represented safe alternatives to phials containing medicinal products of the type used in the typical failure scenario described in Section 3.1.1, Fig. 3.3. The choice deliberately addressed safety concerns related to the toxicity of *Akineton Fiale*, to which the testing operators would have been exposed when the tested phials reached their breaking point (see paragraphs on *Test arrangement* and *Test procedure* for clarification, and Section B.1 in Appendix B for information on safety precautions). The assumption at the basis of this test procedure was that *the material of which the test articles were made corresponded to the material of the medicinal glass phials* relevant for the reference mission described in Section 3.1.1. This assumption was supported by previous investigations of pharmaceutical manufacturers in the Italian and European pharmaceutical landscape cite pharmaceutical producers here?. For the purposes of this test, only the mechanical properties of the glass container were of interest.

For the purpose of efficient data analysis, it is important to note that the phials originated from two distinct medicinal batches, designated as Batch A (36 phials) and Batch B (40 phials), provided by different pharmaceutical producers. The initial assumption that all phials share identical geometry and material properties stems from their classification as standardised containers designed to hold 5 ml of solution. Within the pharmaceutical supply chain, glass container manufacturers typically produce and dispatch these standardised phials to pharmaceutical producers, who subsequently fill them with medicinal product and seal them through specific heat treatments. This implies that parameters such as exact wall thickness variation and specific thermal treatment histories cannot be directly measured prior to testing, and the assumption of the standardised geometric and material properties is ultimately proven incorrect by the experimental results presented in the Section 4.2.2.

Test arrangement and procedure
The compression test was performed using an *MTS QTest/10* testing machine, equipped with a 10 kN load cell for force measurement and control software *TestWork*. The test setup is shown in Fig. 4.2. The phial was held vertically by means of a custom 3D-printed PETG adaptor, which was clamped at its



Figure 4.1: Glass phial containing saline solution, annotated with dimensions.

base by the machine's lower clamp. The compression head of the machine, mounted on the moving crosshead, was lowered from above until contact was established with the tip of the phial, after which the compressive load was applied along the phial's longitudinal axis.

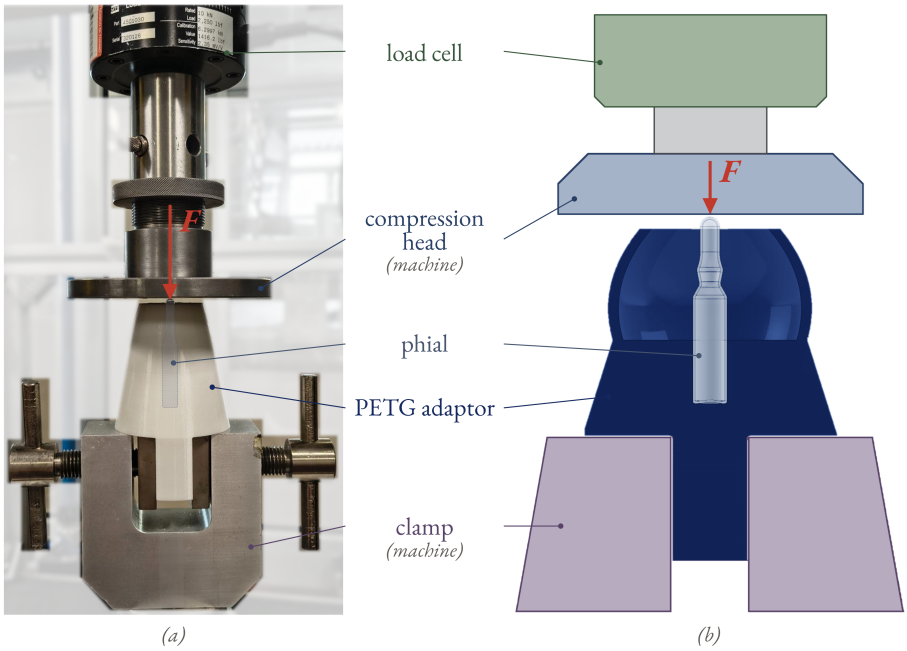


Figure 4.2: Test arrangement: phial held by a PETG adaptor and lower clamp, with the compression head applying the load from above.

The machine was operated under velocity control, measured in mm/min. During the test, the applied force was continuously monitored as a function of crosshead displacement, until phial breakage occurred. Breakage was detected by the testing machine as a sudden drop in the measured load, which triggered an immediate, automatic stop of the crosshead motion.

Test procedure

The failure load was measured at three displacement rates: 1, 2, and 3 mm/min, which guaranteed the quasi-static nature of the test as the measured failure load was observed to be independent of the applied displacement rate (the variation in failure load across the tested rates fell within the expected scatter of the material).

The compression test was repeated on a batch of 76 nominally identical phials (although belonging to two different medicinal batches) to obtain a statistically meaningful population of failure load values. Each phial was visually inspected for pre-existing damage (e.g., cracks or chips) that could affect the test results and was discarded if any defect was found. The phial was then inserted into the PETG adaptor (cleaned from previous phial tests), ensuring correct vertical alignment and full seating within the adaptor up to the height of the phial's main body. Next, the PETG adaptor, with the phial inserted, was positioned within the lower clamp of the testing machine and secured. The compression head was lowered until contact was established with the tip of the phial, taking care not to apply any preload. Finally, the displacement rate was selected, and the test was monitored until phial breakage occurred (triggering an automatic stop of the crosshead). Displacement and applied load data were continuously recorded for each phial.

4.2.2. Results

The medicinal phials under study are not designed as standard coupons to test material properties. In addition, it is important to consider a second complication related to the material of *glass* itself, notoriously characterised by great scatter in mechanical properties. For these two reasons, it is key to introduce some statistical analysis in analysing the experimental results, specifically adopting a two-parameter Weibull distribution parameters to study the phials behaviour. The Weibull distribution is widely recognised in the literature as the most appropriate statistical model for describing the strength of brittle materials like glass due to its ability to capture the weakest-link nature of brittle fracture, which is governed by the size and distribution of surface flaws [70]. Several of the observations were made empirically, i.e. just by looking at the breaking modes or even listening to sound cues in the laboratory, as the almost artisanal setup comes with strong limitations.

Phial breaking modes and statistical observations

Prior to conducting the experimental campaign on the glass phials, it was hypothesised that the phials would break in five areas of potential stress concentration along their geometry. Fig 4.3 illustrates these critical areas: the phial's dome *A*, the first bulge *B*, the necking *C* (the failure point designed to open the phial), the base *D*, and the main body wall *E*. During the tests, it was noticed that not all breaking modes would manifest and that a significant portion of the broken samples would show a combination of two or three failure modes, described as *hybrid* failure modes from this point onwards. Fig 4.3 is a qualitative diagram of various failure modes and exemplifies their nature through pictures taken throughout the testing campaign to accompany empirical observations.

A quantitative synthesis of the data gathered experimentally about phial failure trends can be seen in Fig. 4.4, which categorises the 76 tested phials depending on the observed failure mode. In addition to stiffness and failure load ranges observed for the individual modes, the table in Fig. 4.4 also gives some insights about the correlation between the parent medicinal batch (A or B) and the failure mechanisms. For example, the first row of the table shows that among all the 76 tested phials, 27 of them broke in mode A (dome). Within these 27 phials, 70% belonged to Batch A, and the remaining 30% to Batch B ("Batch mix" column). Furthermore, 53% of all phials of Batch A broke in failure mode A, while only 20% of Batch B showed the same failure mode ("% of batch with this mode" column). Looking instead at the second row of the table, it is possible to see that 18 phials broke in mode E (main body), 89% of them belonging to medicinal Batch B and representing 40% of the entire batch. These two rows give an important insight regarding the *design* of the two medicinal batches. Although nominally identical, batches A and B show different typical failure modes, where batch A favours mode A, and Batch B favours mode E. Generalising this line of reasoning, geometrically identical (as supposedly standard) phials coming from different pharmaceutical producers show radically different mechanical behaviour.

Diverging phial behaviour can be retraced to their manufacturing processes. The phials are originally manufactured open, featuring an aperture in place of the dome to allow them to be filled with the medicinal product. At a later stage, the upper part of the glass is melted and reshaped into the dome. Depending on the medical content, varying levels of control are implemented into the thermal treatment of the phial, as the primary requirements are functionality (breaking in the designed necking when opening the phial) and transportability (complying with packaging and transportation standards CITE STANDARD HERE), over process control. Therefore, it is entirely acceptable for different phials to exhibit distinct mechanical behaviours, even under identical loading conditions and nominally identical

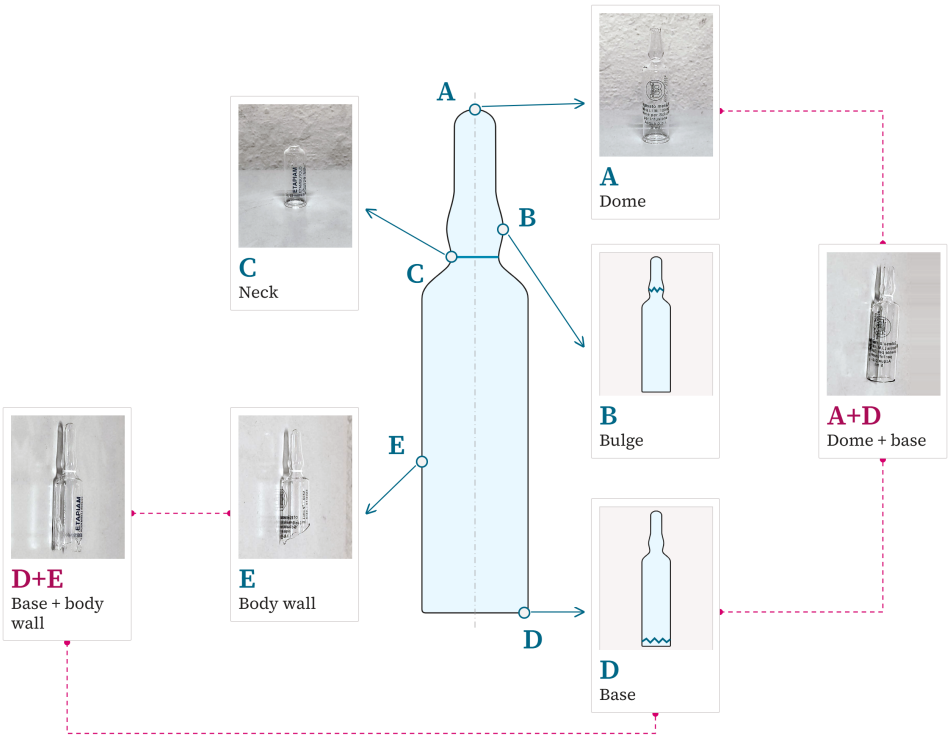
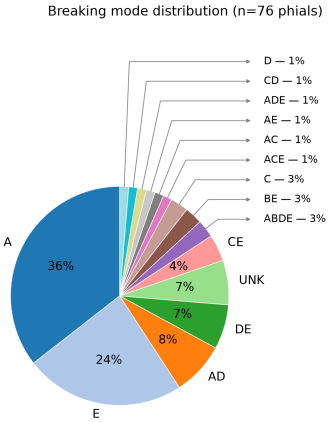


Figure 4.3: Observed phial breaking modes: *Dome* (A), *Bulge* (B), *Neck* (C), *Base* (D) and *Body Wall* (E), together with examples of hybrid modes (A+D and D+E).

geometries, at the cost of needing to differentiate between pharmaceutical batches to draw any significant conclusions on material mechanics.



Mode	n	Stiffness range (N/mm)	Failure load range (N)	Batch mix (of this mode)	% of batch with this mode
A	27	318-1506 (n=26)	61-497	A: 70%, B: 30%	A: 53%, B: 20%
E	18	675-2007 (n=17)	37-933	A: 11%, B: 89%	A: 6%, B: 40%
AD	6	259-1321 (n=6)	140-516	A: 83%, B: 17%	A: 14%, B: 2%
DE	5	940-1698 (n=4)	306-915	A: 40%, B: 60%	A: 6%, B: 8%
UNK	5	506-2107 (n=5)	208-1088	A: 40%, B: 60%	A: 6%, B: 8%
CE	3	1304-2239 (n=3)	518-972	B: 100%	A: 0%, B: 8%
ABDE	2	565-722 (n=2)	389-458	A: 100%	A: 6%, B: 0%
BE	2	271-1254 (n=2)	127-598	A: 100%	A: 6%, B: 0%
C	2	890-1726 (n=2)	442-712	B: 100%	A: 0%, B: 5%
ACE	1	587-587 (n=1)	232-232	B: 100%	A: 0%, B: 2%
AC	1	1200-1200 (n=1)	484-484	B: 100%	A: 0%, B: 2%
AE	1	n/a	133-133	B: 100%	A: 0%, B: 2%
ADE	1	1079-1079 (n=1)	602-602	A: 100%	A: 3%, B: 0%
CD	1	1685-1685 (n=1)	607-607	B: 100%	A: 0%, B: 2%
D	1	245-245 (n=1)	239-239	A: 100%	A: 3%, B: 0%

Mode codes: A = Dome B = Bulging C = Indentation D = Base E = Cylinder | combined letters (e.g. "AD") = multiple modes on the same phial | UNK = mode not identified

Figure 4.4: Distribution of the observed breaking modes.

Stiffness data - glass phials and PETG adaptor

Out of the total 76 phials tested, only 65 phials were considered for the present stiffness analysis, with the remainder omitted due to unprocessability of the gathered data. Load-displacement curves were extracted from the raw dataset to construct the stiffness plot shown in Fig.4.5. To isolate the elastic part of the structural response, data recorded after the first breaking point was discarded for each phial, as the load-bearing capacity of the phials became compromised beyond this threshold. Moreover, knowing that the primary objective of the investigation is to characterise the phials failure load to avoid medical content dispersion during a phial drop event, it is acceptable to deem post-failure behaviour irrelevant.

Categorising the results by batch reveals that Batch B exhibits a visually distinct, generally higher stiffness compared to Batch A, reinforcing the observation that different batches display significantly different mechanical behaviour. Given that thermal treatments typically do not influence stiffness, this discrepancy suggests potential diversity in geometry, disproving the assumption made in the *Test article* section presented earlier. Specifically, a thicker wall in Batch B's phials would account for both higher stiffness and an increased failure load, assuming a comparable failure stress under the current compressive loading state.

Across all curves, an initial compliant branch is observed. This could be attributable to experimental setup constraints as the phials settle and slide into proper contact before effective load transfer occurs. The response remains predominantly elastic across the observed stiffness range, with minor softening observed around 800 N, which is associated with the stiffness compliance of the PETG adaptor (a similar "step" is observed in Fig. 4.6 at comparable loads, indicating some sort of elastic instability in the response of the adaptor). Moreover, a degree of random error must be considered regarding perfect alignment within the PETG adaptor. Deviations from perfect verticality (caused either by clearance between the adaptor and the phial or by subtle manufacturing phial shape imperfections) have a significant influence on the measured stiffness. This aspect is further investigated in the FEM model presented in Section 4.3, which proposes considerations on initial geometric defects to evaluate their impact on overall phial stiffness.

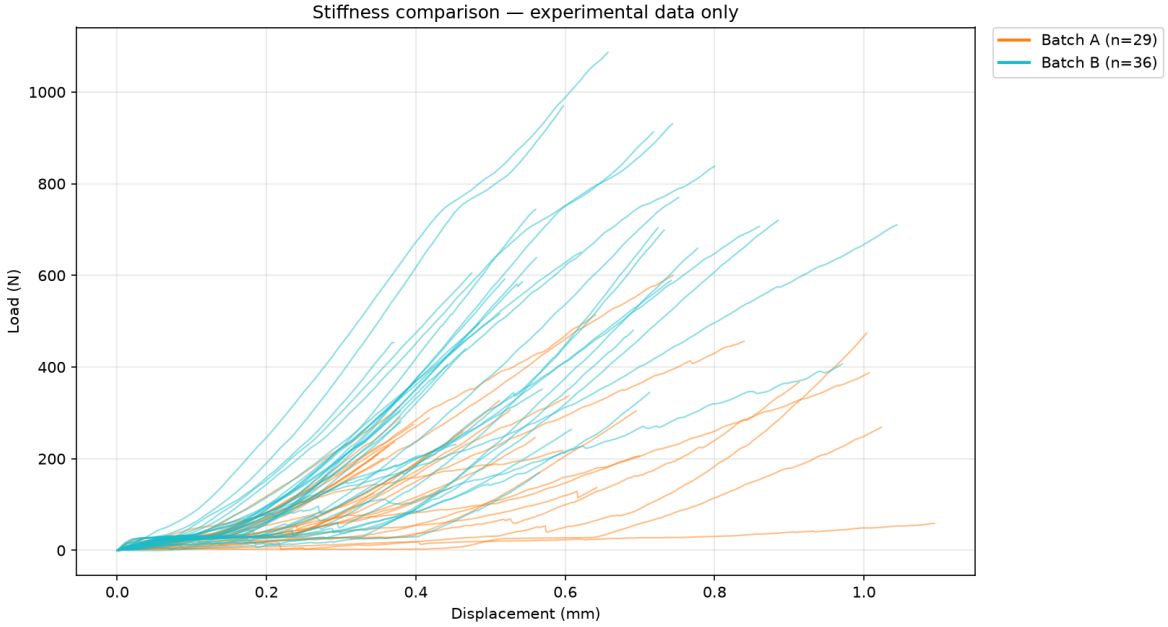


Figure 4.5: Experimental load-displacement curves of all tested phials up to the failure point, grouped by batch (Batch A, $n = 29$; Batch B, $n = 36$).

In retrospective of completing the present experimental campaign and its FEM modelling in Section 4.3, further attention had to be given to the strong limitations of the experimental setup. To validate the stiffness data gathered, it was necessary to conduct a separate investigation on the mechanical compliance of the setup, to determine experimentally the stiffness of the PETG adaptor. For this scope, the adaptor was subjected to the same loading configuration used for the glass phials (refer to Fig. 4.2 for clarification), using a 15cm steel bar of comparable diameter to the glass phials (considered infinitely rigid compared to the PETG structure) to compress the component directly without the risk of inducing instabilities or buckling. The PETG adaptor's load-displacement data is shown in Fig. 4.6, which evaluates the adaptor stiffness at a nominal value of 2037 N/mm obtained by a linear fit within the load band of Mode a failures. Furthermore, it is possible to notice that slight softening behaviour occurred at approximately 800 N, introducing a step-like feature that matches the response observed in the high-load regime of the glass phial tests.

Overall, PETG adaptor stiffness serves as a critical datum for reconciling experimental stiffness with

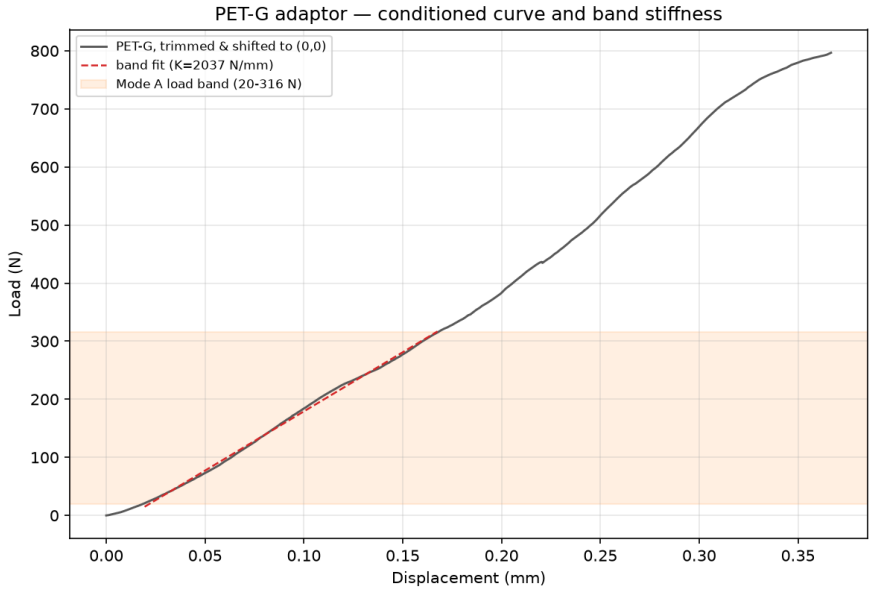


Figure 4.6: Load-displacement curve of the PETG adaptor, trimmed and shifted to the origin. The adaptor stiffness, K_{PETG} = 2037 N/mm, is obtained by a linear fit within the load band of Mode A failures (20-316 N).

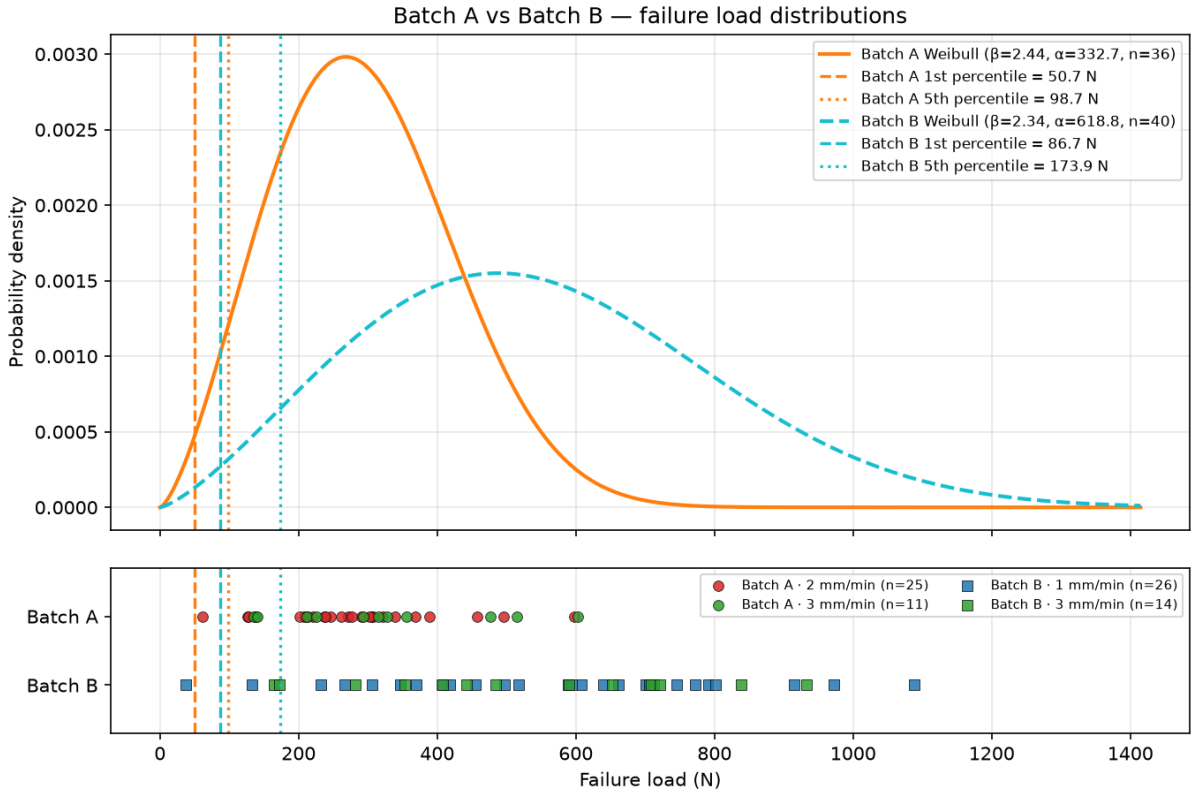


Figure 4.7: Failure load distributions of Batch A and Batch B. Top: fitted two-parameter Weibull probability density functions, with the 1st and 5th percentiles marked. Bottom: individual failure loads, grouped by batch and displacement rate.

numerical predictions of the FEM model presented later. Notably, calculating the adaptor stiffness was not originally planned during the test campaign, but rather it was performed post-hoc to resolve a significant mismatch observed between the empirical stiffness measurements and the initial numerical predictions.

Failure load extraction

Figure 4.7 displays the brittle failure points extracted from the experimental load-displacement curves for all 76 tested phials, categorised by batch and fitted to a two-parameter Weibull distribution. The top plot illustrates the fitted two-parameter Weibull probability density functions, highlighting the 1st and 5th percentiles of each curve, while the bottom plot details the individual failure loads grouped by batch and displacement rate. An analysis of these Weibull distribution plots indicates that while clear stiffness data could not be extracted for every sample, a definitive failure load was successfully obtained across the entire dataset of 76 phials. These failure loads were plotted collectively, regardless of the observed failure mode, to provide an objective picture of the failure loads independent of the manual, visual classification of individual failure modes.

The resulting Weibull distribution highlights that Batch A exhibits a lower characteristic failure load but lower scatter, whereas Batch B shows a higher mean failure load alongside higher dispersion. Consequently, Batch A offers greater reliability for statistical and mechanical analysis and is therefore the only batch considered for the analysis from this point forward. Furthermore, these plots confirm that the testing rate had a negligible effect on the results, as evidenced by the lack of correlation between experimental outcomes and load application speed.

Step 1 deliverable: design failure load of the phial

Based on the considerations made previously, the summary dataset presented in Table 4.2 is selected as the baseline datum to feed into the subsequent stage of the airbag design workflow.

Table 4.2: Experimental data selected for the calibration of the FEM glass material.

Quantity	Value	Source / note
Target failure load (F_{fail})	98.7 N	Batch A, Weibull 5th percentile ($\beta = 2.44$, $\alpha = 332.7$ N, $n = 36$)
Mode A stiffness	318 N/mm - 1506 N/mm	Dome failure; 53 % of Batch A
Mode AD stiffness	259 N/mm - 1321 N/mm	Dome + base failure; 14 % of Batch A
PETG adaptor stiffness	2037 N/mm	Linear fit, 20 N-316 N band (Fig. 4.6)

4.3. Step 2 - Axial compression test (FEM)

INPUT from previous step in workflow vs. OUTPUT for next step

The quantity of interest in the calibration is the tensile strength `FT` of the glass material card in LS-DYNA (`*MAT_GLASS`), which governs the failure force predicted by the FEM model. `FT` is tuned iteratively until the failure force predicted by the FEM matches the value measured experimentally.

The calibrated material model is then used to characterise the glass in a separate FEM model, aimed at predicting the maximum acceleration that the phials can withstand. This value is one of the key design parameters for sizing the airbag protection device.

The current modelling choices are the result of several model iterations, carried out by trial and error, from a first implicit analysis in Abaqus to the present LS-DYNA model. These iterations are summarised in Appendix C.

4.3.1. Implementation

Model setup

explicit analysis MENTION THIS!!! The model is composed of three parts: the base, the crosshead and the glass phial. The idea is to imitate the experimental test, where the glass phial is compressed in the axial direction, i.e. the vertical (Y) direction.

The crosshead and the base belong to the same super-part, called machine, and they are meshed in exactly the same way. Both are made of solid hexa elements with an average size of 0.2 mm, extruded to a thickness of 1 mm in the vertical direction. In plan they measure 18 x 18 mm (base) and 14 x 14 mm (crosshead). Both have enough surface area to cover the base and the dome of the glass phial, so that the breaking pressure and the reaction force from the base are applied correctly.

They are modelled with a material that treats them as infinitely rigid bodies (keyword? e.g. `*MAT_RIGID`), free to move only in one controlled direction, in this case Y. The base is then constrained in Y by a boundary condition that imposes a displacement of 0. The crosshead moves downwards and applies the pressure on the phial in a quasi-static regime. The trick of leaving the Y direction free and then imposing a zero-displacement boundary condition on the base makes it possible to measure the reaction force on the base. With a fully constrained rigid body, this measurement would be harder and would need a slightly different procedure.

The phial is meshed with shell elements of thickness 0.4 mm. The mesh has a regular size (element size?) and is mostly quads, with trias where needed to follow the shape, so that it keeps the curvatures of the dome and the fillets of the rounded-corner geometry. The material used for the glass phial is:

The phial simply rests on the base surface. Its interaction with the base and the crosshead is described by `*CONTACT_AUTOMATIC_SINGLE_SURFACE_ID`, the most general contact model in LS-DYNA. It puts "everyone against everyone" without defining master-slave pairs. The tangential behaviour is defined by a friction coefficient of 0.3 for both static (`FS`) and dynamic (`FD`) friction. The movement of the phial is not restricted in any way. It only feels the reaction force from below and the force applied by the moving crosshead.

The movement of the crosshead is controlled in velocity. It is applied with an initial ramp that scales up to a constant velocity (value?). JUSTIFY RAMP WITH NON-ZERO ACCELERATION AND RISK OF HAVING UNELEGANT SPIKES IN FEM The monitored outputs are the displacement of the crosshead, the displacement of the glass node on top of the dome, and the reaction forces on both the crosshead and the base. In a quasi-static application these two forces must be equal. The application velocity is higher than in the actual experiment, which would be too slow to simulate (too slow to simulate in what sense? runtime?), but still low enough to stay out of the dynamic regime, as explained in the modal analysis section.

native timestep estimation

Modal analysis

A modal analysis was run on the single phial geometry to obtain its natural frequencies and mode shapes, i.e. the eigenvalues and eigenvectors of the free structure. The first 30 modes were requested. The first six are rigid-body modes, since the phial is unconstrained.

The first elastic mode is a bending mode, with a frequency of about 8 kHz. Its period, $T_1 = 1/f_1 \approx 125\ \mu$s, is the characteristic time of the structure's dynamic response. If the load were applied over a time of the same order as T_1, inertia would contribute to the response and the test would fall in the dynamic regime. To keep the simulation quasi-static, the load application time must be at least ten times T_1, i.e. above about 1.25 ms.

For this reason the crosshead is driven by a velocity curve that ramps up over 10 ms to a constant value of [velocity?] and then holds it. The ramp time is therefore about 80 times the period of the first elastic mode, well inside the quasi-static range.

Some of the mode shapes are shown in Figure **??**. They were chosen because their bending deformation is relevant to the phial's failure. These shapes were then used to introduce geometric imperfections into the model, as presented in the next section.

Geometric imperfections

The glass of the phials shows a large dispersion in its material properties, and the geometry itself is only loosely controlled because of the manufacturing process [process?]. For these reasons it was necessary to introduce imperfections into the model, using the geometry perturbation command [keyword?]. Two options were considered:

1. perturbing the nodes along a bending or torsional mode shape from the modal analysis, so as to introduce a specific defect of controlled amplitude [amplitude?] over the nominal geometry;
2. leaving the geometry unchanged and rotating the phial by 1° about its centroid [about which axis?], which introduces a misalignment between the phial axis and the loading direction.

The second option was chosen. Through a trial-and-error process, it was found that this misalignment triggers an additional, hybrid breaking mode, which had also been observed experimentally. The results presented later therefore suggest a correlation between initial deviations from the nominal configuration and the different breaking modes observed in the tests.

Requested outputs

To obtain results comparable with the experimental campaign, a force-displacement curve was extracted from the simulation, using the reaction force on the crosshead and the displacement of the glass node at the top of the phial dome. The objective is to reproduce a brittle failure, in which the glass behaves linearly elastically up to fracture, with no plastic deformation before it breaks.

Calibration procedure

The material card is based on the tensile strength FT, which is used as a calibration knob to shift the breaking point of the phial. The objective is to reproduce in the FEM model the behaviour observed experimentally, i.e. to make the phial break in the simulation at the same load at which it breaks in the experiment. The experimental breaking force is 98.7 N [mean of how many specimens?], so several simulations were run iteratively with different values of FT until the simulated phial broke at 97.[x] N. The resulting value of FT should therefore be read as a calibrated model parameter rather than as the actual tensile strength of the glass, since it also absorbs the effects of mesh size, imperfections and misalignment. This calibrated material model characterises the phial and is used as input to the next step of the workflow.

4.3.2. Results

Quasi-static verification

Stiffness verification

Imperfection selection

FT *calibration*

Comparison with experiments

Step 2 deliverable: calibrated glass material card

LS-DYNA input 4.1: Calibrated glass material card

```
*MAT_GLASS
$HMNAME MATERIAL         5glass_shell
$---+----1----+----2----+----3----+----4----+----5----+----6----+----7----+----8
$      MID       RHO         E        PR                           IMOD      ILAW
          52.6000E-06      70.0      0.23                                      0.0
$     FMOD        FT        FC        AT        BT        AC        BC     FTSCL
        0.0     0.060       1.0                                              1.2
$    SFSTI     SFSTR      CRIN     ECRCL     NCYCR      NIPF
       0.01      0.01       0.0                50.0       5.0
$    EPSCR    ENGCRT    RADCRT    RATENL    RFILTF    FRACEN    CTRACK
      0.100                                                         1
```

4.4. Step 3 - Sled test (FEM)

4.4.1. Implementation

Model setup

Since the geometry of the payload bay was never defined [patented? designed?] by the research group, the shape assumed here is taken from a previous thesis **[key?]** (2001). The model is composed of four parts: the outer shell of the payload bay, the inner sponge, the mask holding the phials, and the glass phials, which represent the medical payload carried in typical missions. The setup is meant to represent a typical carry-on configuration [check wording].

The sponge represents a protective layer such as bubble wrap, of the kind an operator would use to attenuate impacts. Its presence strongly affects the results, since it is part of the attenuation mechanism protecting the phials. Some simulations were also run without the sponge, and in that case the allowable acceleration drops from [value?] to [value?].

Apart from the glass of the phials, none of the materials has a failure criterion implemented. The aim of the analysis is to preserve the phials, not to study the failure of the whole payload, so the payload bay, the sponge and the mask are assumed to remain intact. The model therefore uses three non-glass materials, described below.

The outer shell of the payload bay is meshed with shell elements of 5 mm size and 2 mm thickness, with the material card in Table **??**. The mask holding the glass phials is meshed with shell elements of [thickness?] mm thickness, with the material card in Table **??**. The sponge is meshed with [element type and size?]. Its loading curve was built to describe the typical behaviour of a porous foam: an initial soft branch, followed by hardening as the porosity closes and the material densifies. The curve is based on [source?] and is shown in Figure **??**, with the material card in Table **??**.

The parts interact through `CONTACT_AUTOMATIC_GENERAL` in LS-DYNA, which again treats every part as potentially in contact with every other part. In addition, an interior contact was defined for the sponge to prevent element inversion under large compression.

A reduced model [half? quarter?] of the system was initially developed, but the full model was adopted instead because of the asymmetry of the system [which asymmetry? phial layout? loading?].

Loading

The input of the model is an acceleration time history, prescribed through `*BOUNDARY_PRESCRIBED_MOTION_RIGID` on two rigid rings attached to the payload bay [how are they connected? shared nodes? constrained nodes?], while the rest of the bay remains deformable. The rings represent the straps that hold the payload, in the same way as the restraints on a test sled. The curve has a trapezoidal shape: an initial ramp of duration t_r, a plateau at constant acceleration a of duration t_p, and a descending ramp of duration t_r. The ramps avoid discontinuities in the acceleration, which would excite spurious high-frequency oscillations, so that the velocity and displacement histories remain smooth. The acceleration is therefore applied at the attachment points, and reaches the phials only after being transmitted through the deformable bay, the sponge and the mask.

The impact is characterised by an initial velocity $\mathbf{v}_0$, which includes both horizontal and vertical components [how are the x, y, z components handled?]. Over the deceleration this velocity must be brought completely to zero, so the integral of the acceleration curve must equal the velocity dissipated by the system:

$$\int_0^T a(t)\,\mathrm{d}t = a\,(t_r + t_p) = \|\mathbf{v}_0\|, \qquad T = 2t_r + t_p. \tag{4.3}$$

This condition defines a family of acceleration curves: for a given amplitude a, Eq. (4.3) fixes the plateau duration $t_p = \|\mathbf{v}_0\|/a - t_r$, and therefore the characteristic time scale of the event. The aim is to reproduce an impact as closely as possible, applying an acceleration profile similar to the peak observed in an impact event.

The maximum acceleration the phials can withstand is found with a search procedure analogous to the calibration of `FT`: the amplitude a is increased step by step, with the duration adjusted through Eq. (4.3), until the phial breaks. Since a larger amplitude also implies a shorter pulse, the resulting limit refers to this pulse shape and to the given initial velocity.

The limit is expressed in terms of the acceleration prescribed at the rings [or: measured at ...?]. Acceleration applied in the x, y, and z direction, and sweeping to detect the most critical direction.

Mass scaling

- **Why it's needed.** The 0.1 mm phial shells set the stable timestep ($\Delta t \approx l_e/c \approx 1.9 \times 10^{-5}$ ms). Multiply that by the event duration to show the cycle count is impractical without scaling, even on DelftBlue.

- **How it's applied.** `*CONTROL_TIMESTEP`, `DT2MS`: give the target Δt, and say whether scaling applies only to elements below the target (negative `DT2MS`).
- **Where the mass lands.** Almost entirely on the phial, since it has the smallest elements. Report it as numbers: phial 3 g → 170 g (×57), and the added mass as a % of the total model.
- **Physical consequence.** At a given bay acceleration, the phial's inertial load scales with its mass. The phial–sponge frequency drops by $\sqrt{57} \approx 7.5\times$.
- **How the bias is handled.** Either a native-mass reference run, or the analytical SDOF correction with its validity condition stated (deep isolation, $r \gg 1$, including the native case).
- **Check.** Added mass reported from `d3hsp/GLSTAT`; KE/IE stays low during the plateau.

Requested outputs and failure detection

- **Definition of "broken".** State which event counts as failure: first crack initiation at any integration point (`HISV#1` ≥ 1), or through-thickness failure (all `NIPF` = 5 points cracked).
- **Consistency with Step 2.** The failure event must be the same one used to read the failure force in the `FT` calibration (first force drop). If the definitions differ, `FT` and the allowable refer to different events.
- **Primary indicator.** `HISV#1` crack flag from d3plot, using the max-IP fringe.
- **Secondary confirmation.** Phial internal-energy discontinuity (`MATSUM`), plus the jump in the phial node's acceleration (`NODOUT`).
- **What is not a criterion.** `EPSCR` = 0.1 deletion is a numerical backstop, not a failure definition.
- **Spurious events excluded.** `HISV#1` = −1 (compressive flag), and contact-penetration spikes.
- **Time resolution.** The output interval has to be fine enough to place the crack within the acceleration plateau.
- **Allowable definition.** The highest acceleration level without failure, i.e. the bracket between the last pass and the first fail, with the sweep step stated as the resolution.

4.4.2. Results

Model verification
Acceleration sweep
Failure location and mode
Mass-scaling effect
From sled to impact allowable
Step 3 deliverable: allowable payload acceleration

4.5. Step 4 - Design variables optimisation

4.5.1. Implementation

Problem formulation

* codice ispirato da Zhou et al 2019 - spiegazione del modello e dell'aspetto dei vari blocchi * Sapendo che le thresholds delle provette sono tot, allora si può convergere verso un pre-dimensionamento dell'airbag con tot dimensioni o caratteristiche. * oracle code for direzionamento verso un punto di partenza per il design

* I CHOSE THE CRUDE IMPLEMENTATION — THIS IS BECAUSE THE ONLY COUPLING ANALYTICAL MODEL IS MISSING AND I CANNOT RETRIEVE IT. THEREFORE THIS IS MY IMPLEMENTATION XXX Title: "Design, Fabrication and Testing of an Airdrop Platform Utilizing Airbags as Shock Absorbers"

Author: Timothy F. Patterson

Report: NATICK/TR-87/042 (DTIC accession ADA199593)

Date: August 1985 (work performed March 1984–May 1985)

Physical model

```
init: geometry(h=0), m0 = P0·V0/(R·T0)
loop while t < t_max:
  choose dt from previous load (hysteresis)
  F_n = max((P-Pamb)·A_prev, 0) → a_x, a_y          [explicit: old P, old A]
  u_new = u + a·dt ;  h_new = clamp(h + u_n,new·dt)  [semi-implicit Euler]
  A, V = geometry(h_new) ;  A_vent = A_or + φ(S - A)
  ρ = m/V → P, T adiabatic from previous state
  repeat N sub-steps (dt/N):                          [sub-cycling, not iteration]
      if P ≤ Pamb: stop
      dm = ṁ(P,T,A_vent)·dt/N → m, ρ, P (old T), T
  P = max(P, Pamb)
  if P > P_burst → BURST  | if h ≥ D0 → BOTTOMED | if y_pen ≤ 0 → REBOUND
  advance state
```

Optimisation algorithm

Table 4.3: Inputs of the preliminary airbag optimisation: fixed inputs and boundary conditions (left) and optimisable design variables (right).

Fixed inputs and boundary conditions		Optimisable design variables	
Impact scenario		**Continuous (searched by the optimiser)**	
M_p	0.5 kg, payload mass	D_0	0.39 m to 1.5 m, diameter
u_0	17.2 m s^{-1}, vertical impact speed	L_0/D_0	0.5 to 4.0, aspect ratio (cylinder only)
$u_{x,0}$	5.0 m s^{-1}, horizontal impact speed	P_0	P_{amb} to 1.4×10^5 Pa, inflation pressure
$u_{n,0}$	≈ 17.9 m s^{-1} *along* $\mathbf{n}$, *at* $\theta \approx 16.2°$ *(derived)*	φ_{vent}	0 to 0.10, vent open-area fraction
Environment		**Discrete – integer (searched by the optimiser)**	
P_{amb}	101 325 Pa, ambient pressure	n_{ply}	$\{1, \dots, 4\}$, number of fabric plies
g	9.81 m s^{-2}	**Discrete – categorical (enumerated)**	
T_0	288 K, initial gas temperature	Shape	sphere ($k = 0.5$), cylinder ($k = 1$)
Requirements and safety factors		Fabric	Nylon 6,6, silicone-coated: 600 N cm^{-1}, 180 g m^{-2}
a_{allow}	500 m s^{-2}, allowable payload acceleration		PET, PU-coated: 550 N cm^{-1}, 190 g m^{-2}
SF	1.2, safety factor on a_{allow}		Aramid, coated: 1000 N cm^{-1}, 260 g m^{-2}
a_{lim}	$a_{allow}/SF \approx 417$ m s^{-2} *(derived)*	Gas	N_2: $R = 296.8$ J kg^{-1} K^{-1}, $\gamma = 1.400$
$u_{res,max}$	2.0 m s^{-1}, residual speed cap		He: $R = 2076.9$ J kg^{-1} K^{-1}, $\gamma = 1.667$
η_{seam}	0.70, seam efficiency on fabric strength		Ar: $R = 208.1$ J kg^{-1} K^{-1}, $\gamma = 1.667$
SF_{burst}	1.50, design factor on membrane tension		CO_2: $R = 188.9$ J kg^{-1} K^{-1}, $\gamma = 1.289$
PEN	10^4, penalty weight on constraint violations		Dry air: $R = 287.0$ J kg^{-1} K^{-1}, $\gamma = 1.400$

4.5.2. Results

Reference optimum
Convergence of the optimiser
Effect of the 2D impact
Effect of transpiration and seam efficiency
Sensitivity to the inherited inputs
Step 4 deliverable: airbag sizing parameters

4.6. Summary of results

RESULTS: BRING IT BACK TO THE RESEARCH QUESTIONS!!!!!

idea: weighted payload to predict exactly where it will fall, airbag does not need to wrap around the entire payload but rather only the part that is guaranteed impact. This saves material and improves chances of not breaking the payload, if it actually falls in a predictable asset.

4.7. System design hypothesis

5

Conclusions

5.1. General summary

Bring back the idea of the safety-first workflow into the broader context of the long-term, drone-network-all-over-Italy vision.

5.2. Recommendations for future work

* practical: go to designing, testing, etc. * theoretical: improve workflow with reliability spin (as when i wanted to change my thesis), to improve airbag design and possibly save on mass still * idea of parachute-airbag optimisation, where parachute slows down initial descent, then payload is sganciato and airbag comes into action ?

Feeding these results in the "outer" design loop, updating the hardware and mission design, re-evaluating initial conditions to re-inform the "inner" design loop -> once reached convergence, the need to simulate and test the design implemented on the drone, with informed decisions on inflation mechanism and folding method.

combination of parachute (for, let's say, half the falling height) and then airbag for the second half? the objective is to minimise oveall mass.

we are not done at all. There is the geometry-specific part that is required, including the design recommendations in Section 2.3.2 and going on, PAST the workflow, as mentioned in Section 3.2.2

6

Project Management

6.1. Project organisation and stakeholders

6.1.1. Context

This thesis was conceived as a continuation of a research line initiated by Politecnico di Torino in 2020, during the COVID-19 emergency, and carried on until 2023 with the last master's thesis before this one. The former theses [five?] worked at a conceptual and evaluation level, with no hardware, focusing on the delivery of medical goods in the Turin territory through UAV-based operations. This is the first of these theses to take a design-oriented step, and it formalises the findings of the former ones.

The formal setup is a TU Delft graduation agreement, with Politecnico di Torino as the host university, FEV Italia as the industrial partner and the Molinette hospital as the advisory hospital. The official start was on [23 or 25?] March 2026, with a nominal duration of eight months.

6.1.2. Supervisory team

The supervisory team was composed of five members. First, Bianca Giovanardi from TU Delft was the responsible supervisor of the thesis, followed by a series of external supervisors belonging to three different Italian institutions. The host institution, Politecnico di Torino, provided two supervisors: Prof. Enrico Cestino, the main professor interested in this research line, and Davide Angelini, PhD student and co-supervisor of this thesis. At an industrial level, Eng. Sergio Bianco from FEV Italia S.r.l. advised on the FEM parts of the thesis. Finally, Dr. Daniela Cestino, representative of the Città della Salute and of the Molinette hospital, acted as medical advisor.

Each of the supervisors requested a different means of communication. Structured meetings were organized mostly with the universities, while ad-hoc meetings were organized with S. Bianco and D. Cestino [?], depending on their availability. The frequency of updates and meetings is summarized in Table 6.1.

Table 6.1: Supervisory team, roles, and communication channels.

Supervisor	Organisation	Role	Channel and frequency
B. Giovanardi	TU Delft	Responsible supervisor	Bi-weekly meetings
E. Cestino	Politecnico di Torino	Host supervisor	Weekly meetings, skipped when there was no update
D. Angelini	Politecnico di Torino	Co-supervisor	Weekly meetings, together with E. Cestino
S. Bianco	FEV Italia S.r.l.	Industrial co-supervisor	Ad hoc, fitted to his availability
D. Cestino	A.O.U. Città della Salute	Medical co-supervisor	Ad hoc, for supply of the medicinal phials

6.1.3. Monitoring and communication

The way the individual supervisors were consulted already poses a challenge for how communication was conducted throughout the thesis, i.e. coordination was not straightforward. The only joint meeting among all the relevant parties was the pre-kick-off alignment, attended by [X, Y and Z], while S. Bianco and D. Cestino were only ever consulted individually. As a result, the student was the only information link between the parties involved, and each supervisor saw a partial picture through a separate channel, so expectations diverged over time [example?].

- Progress was reported weekly or bi-weekly to the academic supervisors throughout. Supervisors could follow the modelling difficulties as they happened.
- Reporting was transparent about difficulties as well as progress. Each milestone presentation tracked every work package and design step as planned, in progress or done, including steps not achieved (for example WP3 and Steps 2–3 at the midterm).
- Meetings were adapted to need: weekly sessions were skipped when there was no update, and industrial meetings were fitted to availability.
- **Weakness:** there was no shared progress record across supervisors, and no joint review after kick-off.
- **Weakness:** the change of scope was not recorded in a dated decision.

6.1.4. Stakeholder priorities

The process of aligning the expectations of the different parties also posed a challenge for this thesis, as the home and host institutions had divergent priorities. Politecnico di Torino and FEV Italia were interested in a rigorous characterisation of the glass phials and, in general, of the medical payload. For TU Delft, the main concern was that the scope be feasible within the nominal time [?]: the TU Delft supervisor judged the planned work to be more than sufficient from the start, and at the beginning even overshooting.

6.1.5. Resources by party

The resources offered by the stakeholders are listed below, focusing specifically on what each of them offered and contributed to the thesis.

- **TU Delft:** supervision and access to the DelftBlue HPC.
- **Politecnico di Torino:** the MTS compression testing facility, plus HPC and licences as expected in the research proposal.
- **FEV Italia:** a workstation with HyperWorks, and modelling support.
- **A.O.U. Città della Salute (Molinette):** the medicinal phials.

6.2. Planning: baseline versus execution

6.2.1. Baseline

The timeline hypothesised at the beginning of the MSc Thesis project is shown in 6.2, where the different project phases and milestones presented in the first project proposal are highlighted in chronological order.

6.2.2. Scope evolution and execution

As expected [why?], the thesis work evolved away from the initial planning. The project proposal at the beginning considered the full design, with four sub-questions, including regulations and experimental testing of samples [?], and an end date of 30 September. In the new research proposal [?], the main question became workflow-centred. The scope then changed again as the iterations went by. The main deviation was Step 0 [?], which was planned to take three weeks and took several months, as higher importance was given to the modelling than initially hypothesised. At the midterm review, sub-question 3 was put in brackets, i.e. set aside, and the scope moved to its current state. Although no formal replan was made after the midterm, it was agreed among the supervisors that the focus would be on a thorough [honest?] characterisation of the payload before informing the airbag design, with a

Table 6.2: Baseline planning from the research proposal.

Phase / milestone	Planned duration	Planned date (opt. / max.)
Phase 0: Mission envelope and allowables	3 weeks	19/06
Phase 1: Inflated configuration	7 weeks	07/08
Midterm review	–	31/07 - 07/08
Phase 2: Folding and deployment	10 weeks	16/10
Green light review	–	23/10 - 13/11
Final submission	–	13/11 - 11/12
Phase 3: Refinement for prototyping	*Future work*	–

general readjustment of the timeline to allow for the regular conduction of the green light meeting and the delivery of the final portfolio.

- **Project proposal V2:** full design, 4 sub-questions including regulation, experimental testing on pharmaceutical samples, and an end date of 30/09.
- **Research proposal:** the main question becomes workflow-centred; there are 3 sub-questions; the regulatory question is dropped; and experiments become "if required".
- shifing the scope as iterations go by. The main deviation was Phase 0: planned at 3 weeks, it took several months.
  - About 6 model iterations were needed for the compression test.
  - Several more were needed for the sled test, including the sponge.
- **Midterm review (04/08):** sub-question 3 (folding and deployment) is put in brackets.
- The midterm review was held on 04/08, inside its window.
- **After the midterm:** the focus moves to the workflow and characterisation. No formal re-plan was made after the midterm.
- Green light is now expected on 13/11 and submission on 30/11, both inside the windows set in the research proposal. They may move earlier if the work finishes sooner.

6.2.3. Final scope definition

The final evolution of the research process settled in this way: Sub-question 1 (allowables): answered in full, through experimental and FEM characterisation. Sub-question 2 (inflated configuration): answered at pre-design level, using an optimisation wrapper extended to 2D (including horizontal velocity, seam efficiency and fabric transpiration). Sub-question 3 (folding and deployment): moved to future work. Main question (workflow): answered. The upstream steps were developed with high fidelity. Closing deliverable: a design recommendations chapter, concluding that the design must be payload-specific before prototyping. Instead of putting the original research questions, describe them better earlier and explain the evolution here.

6.3. Critical events and their management

6.3.1. Scope narrowing: from airbag design to workflow and characterisation

About two months into the research process, right after the literature research phase, the TU Delft supervisor raised the possibility of narrowing the scope and framed two branches: the thesis would focus either on the airbag design or on the pre-design, i.e. the workflow before the airbag design. The two were practically mutually exclusive. The student first chose the airbag design branch, but the question was not formally presented to the Politecnico di Torino supervisors, whose priority on the phials had been known since the initial definition of the thesis. In hindsight, this is where the divergence of expectations described above started [check].

The second decision point came after the midterm. By then it was clear that building high-fidelity models in the pre-design part of the thesis took far more iterations than planned, together with the

realisation that the downstream sizing depends on the upstream allowables, so sizing on unverified models would carry their errors forward. This was the basis of the reasoning for insisting on the modelling part, and it aligned perfectly with the expectations of Politecnico di Torino and FEV Italia, whose position was "credible allowables first". The outcome was that the scope moved from designing the airbag, as initially planned, to the workflow before the design of the airbag. This meant taking the road the TU Delft supervisor had already outlined at the beginning of the thesis, while keeping the main research question [check against the previous paragraph]. In this case the scopes aligned, and everybody agreed that the design must be payload-specific and that characterisation first was the right order.

6.3.2. Loss of computational resources

Up until the midterm review, which took place at the beginning of August, the only resources available were a local machine running Python and an LS-DYNA student licence, limited to four cores and 128 000 elements [nodes?]. The expected HPC access, which was supposed to be granted about a month before the midterm meeting, was delayed by maintenance. This meant that a further delay was forecast: it was hypothesised that, to unlock new computational power at Politecnico di Torino, the student would need to wait at least another month, with a one-month queue to be expected, for a delay of three months in total, including the month already lost. At that point the options were to continue with the Politecnico di Torino HPC and ask for an extension, to obtain computational power elsewhere, or to change the thesis topic, either salvaging the work or redefining it completely. In the end, computational power was unlocked thanks to the intervention of TU Delft, which offered access to its DelftBlue cluster outside of the thesis agreement, as something not agreed upon before the thesis, and the topic was not changed.

6.3.3. Split toolchain between workstations

Another constraint concerned the toolchain used for pre- and post-processing of the same calculations. The FEV workstation had HyperWorks installed with industrial licences, but the IT policy of the company blocked the installation of the VPN needed to connect to DelftBlue and launch the LS-DYNA input files there. On the contrary, the personal PC had VPN access to DelftBlue but no HyperWorks installed, and the only usable pre- and post-processor was LS-PrePost, which is not nearly as flexible as HyperWorks.

The impact on the project was that data could only flow out of the FEV PC, so the DelftBlue results could never be post-processed in HyperWorks. Every large model change required the full round trip of going back to the FEV PC and editing the model there, and large files risked exceeding the limits of the only transport channel, e-mail [why only e-mail?]. This caused many time-consuming issues and a significant loss of time. It also carried the risk of losing files and of model incompatibilities: parameters were sometimes edited on the personal computer and then not implemented on the large model, so the same changes had to be done again. In general, the lack of transferability of the models was particularly difficult to manage. The small mitigation that was adopted was splitting the model into modular input files (`*INCLUDE`), so that only the parts being changed had to be transferred. This is also standard practice in industrial FEM; however, not having the same pre- and post-processor on the same machine remained a challenge.

6.3.4. Technical decisions under conflicting advice

On two occasions, the decisions of the student conflicted with the advice given by one of the supervisors.

Modelling approach for the phial. The first example concerned the modelling approach for the phial. The industrial supervisor suggested a solid-element model with a material card developed for concrete and ceramics. With this approach, however, there were still contact issues, the material behaviour was not adequate to the type of failure observed in the lab, and the reaction force was zero [noisy?] or strongly oscillating in the model, pointing to contact sensitivities and to a material model not formulated for glass. The student therefore decided to switch to shell elements and, after an independent literature review, to adopt a glass-specific material model, `*MAT_GLASS` (MAT_280). This made it possible, for the first time, to obtain reaction forces that were measurable and comparable with the MTS data.

Fidelity of the sled-test model. The second occasion was when the TU Delft supervisor advised using a reduced model for the sled test, because of the computational cost and the risk of not finishing on time. However, the student realised that the reduced model relied on a symmetry that the system does not have, so its results could not be considered physically valid, nor presented as such to a thesis committee with a critical judgement. The decision was therefore to build the full container model, keeping the reduced model as a fallback plan and taking full responsibility for any delay. As an outcome, the full model was built successfully and the green light date did not move.

6.4. Self-assessment

Self-evaluation of performance is important. On one hand, what worked is Milestones were held despite a forecast delay of about 3 months: the midterm was on time and the green light and submission stay inside the proposal windows; A recommended technical approach was overturned with evidence: the shell elements and glass material model that were adopted produced the first usable comparison with experiment. Somehow adapting to the communication style of every supervisor and carrying the project forward despite the accidents above.

on the other hand, in all honest evaluation, there were also things that did not work. Phase 0 was badly underestimated. 3 weeks were planned against several months actually needed, because the iterations required for brittle-failure and contact modelling were not foreseen. The divergence in supervisor priorities was never raised openly. Trying to deliver both the airbag work and the phial characterisation increased the workload and delayed convergence on scope. Too long was spent on the recommended solid-element approach before switching, because there was no criterion for when to stop.

IDEA: propose it as the natural evolution of research, you start with one thing and then understand that much more work is necessary to actually do it.

6.5. Lessons learned and recommendations

Overall, the lessons learned from this magical experience were:

- Schedule joint milestone reviews (kick-off, midterm, green light) with all supervisors in the initial plan.
- Keep one shared progress record that every supervisor can access. This is against what was adopted, i.e. different platforms for different supervisors (github, teams, dropbox, etc.)
- Keep a decision log, and confirm scope changes in writing on the date they are made.
- Put scope conflicts to all parties early and get a decision, rather than absorbing them as extra work.
- Set stopping criteria for technical approaches, for example “after N iterations with no usable output, escalate or switch”.
- Plan iteration buffers for modelling tasks that are new to you, especially brittle failure and contact.
- Confirm resource availability (HPC, licences, VPN, software on each machine) in the agreement before kick-off, not only in the proposal.

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A

Extended Market Analysis

A.1. Value-chain actors

Table A.1: Actors involved in drone delivery of medical goods in Italy, grouped by value-chain stage. † Based in Piedmont.

Category	Institution	Scope
Drone manufacturers	SkyEU†	Autonomous high-speed eVTOL drones for first aid and critical logistics [27]
	FlyingBasket	Cargo drones; used for blood transport Pozzuoli–Ischia [71]
	H2C	Hydrogen drones and hydrogen supply system for Padova Sandbox [72]
	Dronus	"Radon" drone for medical delivery [73]
	Eurolink Systems	"Beluga" medical delivery drone, with Università Cusano [73]
	IDS Ingegneria Dei Sistemi	Defibrillator-carrying drone (WingBeat project) [74]
Technology partners (role not specified)	Mavtech	INDOOR technical partner; aerial surveillance and operational-support products [18]
	FEV Italia†	Industrial partner of the DIMEAS feasibility studies; numerical impact analysis
	Leonardo	VTOL drone demonstration flights in Lazio, with Telespazio and D-Flight [75]
	Caltec	Technical partner of SIS 118 for the SEUAM defibrillator drone [76]
	AP 74 UAV	Drone company involved in the Piadena pharmacy link [77]
Payload / container makers	ABzero	Smart Capsule: sensorised, UN3373-certified container for blood, organs and drugs [21]
	Carpitech	"Philotea" temperature- and atmosphere-controlled capsule [78]
	Air Abruzzo	"Flying Box" transport capsule [78]
	Telespazio	TPZ Air 100 drug box used in U-ELCOME [79]
Operators	ABzero	Drone operator for San Raffaele trials; handled ENAC authorisation for INDOOR [23], [80]
	Unmanned4you	Operated Aetos blood-transport flights with FlyingBasket [71]
	DL Droni	Piloted the Piadena pharmacy flights [77]
	Telespazio	Runs the U-ELCOME hospital-to-village drug delivery service [16]
	Croce Rossa Italiana	Own drone fleet and certified pilot school; urgent-drug delivery trials [81], [82]
Airspace, infrastructure and energy	ENAV	Air navigation service provider; coordinated Corus-XUAM [83]
	D-Flight	U-space services (ENAV group); Corus-XUAM partner [83]
	DTA	Corus-XUAM partner, Grottaglie airport test site [83]

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Category	Institution	Scope
	ADPM Drones	"Hangar": docking and recharging station for automated missions [83]
	Gruppo SAVE	Airport group leading Padova Sandbox [72]
	Heron Air	Operator of Padova airport, Sandbox partner [72]
	Milani; DBA Group	Build the hydrogen production and refuelling unit for Padova [84]
	Stesi	Supervision system, ground logistics and spare-parts analytics for Sandbox [85]
	Torino Caselle and Cuneo Levaldigi airports†	Proposed hub and support base for hydrogen drone logistics [24]
Research	PIC4SeR, Politecnico di Torino†	Autonomous flight technologies for transporting biological material (INDOOR) [20]
	DIMEAS, Politecnico di Torino (Prof. E. Cestino)†	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost–safety comparison of car, drone and eVTOL, hub siting [9], [10], [11], [12], [13], [14]
	Università di Torino†	INDOOR project partner [18]
	Scuola Superiore Sant'Anna	Parent institution of the ABzero spin-off [23]
	Università Cusano	Co-developer of the "Beluga" drone [73]
	San Raffaele technology centre	Developed the FF2020 drone delivery demonstrations [86]
End users	AOU Città della Salute e della Scienza†	INDOOR test flight between hospital sites; proposed hydrogen logistics plan; pharmacy department partner of DIMEAS feasibility studies [22], [24]
	Regional Transplant Centre Piemonte–Valle d'Aosta†	INDOOR partner for organ and biological transport [18]
	Ospedale San Raffaele	In-campus delivery of drugs and biological samples [87]
	ASST Sette Laghi	Feasibility study for inter-hospital drug transport [88]
	Piadena community pharmacies	Pharmacy-to-pharmacy drone link [77]
	ASL 1 Abruzzo	Hospital-only drugs delivered to remote villages [16]
	ASL Napoli 2	Blood transport to Ischia [71]
	Ospedale Miulli	Demonstration flight of life-saving drugs [89]
	ASL Lecce	Planned Gallipoli–Casarano hospital link [90]
	Ospedale di Patti	Planned link to the Aeolian Islands [90]
	Ospedale Pediatrico Bambino Gesù	Partner in VTOL demonstration flights in Lazio [75]
	SIS 118	Emergency service promoting the SEUAM defibrillator drone [76]
Promoters and funders	Fondazione DOT†	Promoter of the INDOOR project [18]
	Centro Nazionale Trapianti	Funds the INDOOR research grant at PIC4SeR [19]
	City of Turin†	INDOOR partner; funded the LastMed call [18], [25]
	Regione Veneto	Signatory of the Padova Sandbox agreement [72]
	Regione Lazio	Partner in VTOL demonstration flights [75]
	Regione Campania	Institutional support for the Aetos trial [91]
	EU research programmes	H2020 Flying Forward 2020; U-ELCOME, co-funded by Horizon [80], [92]
	Abruzzo municipalities and Parco Sirente Velino	Host communities of the U-ELCOME service [16], [79]
Regulators and authorities	ENAC	National civil aviation authority; flight authorisations; protocol with the Red Cross [93]
	EASA	EU rules for drone operations, Regulation (EU) 2019/947 [94]
	Regione Piemonte†	Regional health authority; drone trials subject to benefit–risk analysis [28]
Alternatives	Tecnocad / Tecnomill / Bylogix†	LastMed autonomous ground rover for last-mile drug delivery [26]

A.2. Trials and planned initiatives

Table A.2: Drone delivery of medical goods in Italy: trials and planned initiatives (status as reported by the cited sources; checked 5 October 2026).

Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
Piedmont	Turin, Molinette ↔ San Giovanni Bosco feasibility studies	2020–2024	DIMEAS (Politecnico di Torino); AOU Città della Salute e della Scienza (Pharmacy); FEV Italia	Life-saving drugs, blood-derived medication	Approx. 10 km corridor along the Po, approx. 15 min at 17 m/s	Desk studies (5 MSc theses, 1 conference paper); no flights performed	[9], [10], [11], [12], [13], [14]
	Turin, INDOOR	Oct 2024	Fondazione DOT; PIC4SeR (Politecnico di Torino); ABzero; AOU Città della Salute e della Scienza	Biological samples (blood) in sensorised capsule	CTO → Molinette, 500 m, automatic navigation over the city and the Po	Single experimental flight	[18], [22]
	Turin, hydrogen drone logistics plan	Apr 2025	AOU Città della Salute e della Scienza (Commissioner T. Schael)	Drugs, small medical loads	Torino Caselle hub, Cuneo Levaldigi support, future Parco della Salute	Proposal; no implementation found	[24], [95]
Abruzzo	Avezzano → Collelongo, U-ELCOME	Jul 2025	Telespazio (Leonardo/Thales); ASL 1 Abruzzo; municipalities of Collelongo, Montereale, Rocca di Mezzo	Hospital-only drugs for frail patients	Hospital to village pharmacy, approx. 20 min	Start of operational phase; further routes announced	[16], [17]
Apulia	Grottaglie → Manduria, Corus-XUAM	by 2022	ENAV; D-Flight; DTA	Medical material	Airport to hospital	Test (reported in 2022)	[83]
	Taranto, SEUAM	Feb 2024	SIS 118; Caltec (Campanian aerospace consortium)	Automated external defibrillator	Urban, simulated cardiac arrest	First test of experimental project	[76]
	Lecce, ASL Lecce	2025	ASL Lecce	Not specified in source	Gallipoli–Casarano hospitals	Intended	[90]
	Acquaviva delle Fonti, Ospedale Miulli	Nov 2025	Ospedale Miulli	Life-saving drugs, biomedical material	25 kg hexacopter; 25 km in 25 min	Operational demonstration	[89]
Campania	Pozzuoli → Ischia, Aetos	Feb 2025	FlyingBasket; Unmanned4you; ASL Napoli 2	Blood, biological samples	17 km in 15 min; 4 flights; 5 kg carried	Second trial phase	[71], [96]
Lazio	Rome	2022	Leonardo; Telespazio; D-Flight; Ospedale Pediatrico Bambino Gesù; Regione Lazio; Aeroporti di Roma	Not specified in source	Fixed-wing VTOL drones	Two demonstration flights	[75]
Lombardy	Piadena (Cremona)	2021	Two community pharmacies; DL Droni; AP 74 UAV	Drugs and devices between pharmacies	Approx. 6 kg electric drone, approx. 1 kg payload	Announced as continuous service; current status not verified	[77]

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Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
	Milan, Ospedale San Raffaele, H2020 Flying Forward 2020	Mar 2023	Ospedale San Raffaele; EuroUSC Italia (regulatory consultant); ABzero (operator)	Drugs, biological samples	Within the hospital campus	Demonstration campaign, each flight ENAC-authorised	[80], [87]
	Varese, ASST Sette Laghi	2025	ASST Sette Laghi	Drugs, urgent materials	Between two hospitals a few hundred metres apart	Preliminary feasibility study	[88]
Sicily	Trapani, Croce Rossa Italiana	2021	Croce Rossa Italiana	Urgent drugs	Port quay → vessel; 5 min 30 s vs 35 min by sea	Simulation exercise	[82]
	Patti → Aeolian Islands	2025	Ospedale di Patti	Not specified in source	Patti–Lipari	Under development	[90]
Veneto	Padova, Padova Sandbox	from Mar 2025	Gruppo SAVE; ENAC; Regione Veneto; H2C/H2G; Heron Air; Milani; DBA Group; Stesi	Medical goods, biological samples (hydrogen drones)	Up to 100 km, 4–5 kg payload (stated)	Ground tests of hydrogen system; first flight expected autumn 2026	[72], [97]

B

Safety Measures for the Experimental Campaign

B.1. Safety precautions

The glass phials used in this test contained saline solution. The possibility of skin or eye contact with the liquid content or glass fragments in case of phial rupture was nonetheless considered. To mitigate the risks associated with both fragment ejection and liquid release, the following precautions were adopted:

- **Precaution 1:** The phial was fully constrained within the PETG adaptor up to the height of its main body, as shown in Fig. 4.2, limiting the lateral dispersion of both glass fragments and liquid content upon failure. A small clearance was left between the top of the phial and the top of the PETG adaptor, sufficient to allow the compression head to apply the load directly onto the phial without interference from the adaptor itself.
- **Precaution 2:** All personnel present in the test area wore safety glasses for eye protection for the entire duration of the test.
- **Precaution 3:** The testing machine was operated under velocity control, with the applied force continuously monitored; upon detection of the load drop associated with phial failure, the machine automatically and immediately halted crosshead motion, preventing any further loading of the fragmented test article.

No other significant hazards (electrical or otherwise) were associated with the test, as the testing machine was operated within its standard, intended use envelope.

B.2. Persons allowed in the test area

Access to the test area during the execution of the test was restricted to the following personnel:

- the author of the present thesis, Annamaria Iannuzzo;
- PhD candidate, MSc Thesis co-supervisor, and test supervisor Davide Angelini;
- MSc Thesis co-supervisor, prof. Enrico Cestino.

No other personnel were permitted within the test area for the entire duration of the test.

C

Development Iterations of the Glass Phial FE Model

Table C.1: Sequence of FEM modelling iterations for the axial compression test.

V	Solver	Model / elements	Material / failure	Outcome / reason for change
1	Abaqus/Standard (implicit, static)	2 ml Akineton phial, quarter model; C3D10 (glass), S4/S4R stiff platens; contact pairs	Elastic + MAXPE damage initiation, linear displacement-based evolution	Contact convergence issues; usable force–displacement curve only in the purely elastic case
2	Abaqus/Explicit	As V1; general contact; variable mass scaling	`*BRITTLE CRACKING/SHEAR/FAILURE`; strain- and energy-based cracking variants	Immediate element erosion, strong oscillations, no readable reaction force. Switch to LS-DYNA (limited Abaqus licences; student licence available) and to the 5 ml saline phial (safety)
3	LS-DYNA explicit	5 ml phial; 10-node tetrahedra (ELFORM 16)	`*MAT_BRITTLE_DAMAGE` (MAT_096), as recommended by the industrial supervisor	No functional breaking criterion; reaction force not readable
4	LS-DYNA explicit	0.1 mm hexahedra; platen 100 mm/s, 2 mm stroke, no mass scaling	MAT_096	Breaking criterion active, but reaction force null or strongly oscillating
5	LS-DYNA implicit	Hexahedral model of V4	Linear elastic; implicit breaking criteria explored	Stiffness far above MTS value; implicit criteria unsatisfactory. Return to explicit
6	LS-DYNA explicit	Shells ELFORM 16, NIP = 5, mid-surface map mesh (thick shells rejected). Quarter model at 0.5 mm first, then full model at 0.1 mm	`*MAT_GLASS` (MAT_280); EPSCR raised from 0.005 to 0.1 after it was found to govern deletion	First measurable reaction force comparable with MTS data. Quarter model dropped: convergence issues at 0.5 mm and imposed symmetric crack pattern; full model enabled by DelftBlue HPC
7	LS-DYNA implicit (eigen)	Full 0.1 mm model of V6	–	Mode 18 (bending-type) selected as deterministic imperfection
8	LS-DYNA explicit (final)	Full model, 0.1 mm shells (batch-A geometry, caliper-measured); mode 18 at 0.08 mm via `*PERTURBATION_NODE` + 1° tilt about horizontal centroidal axis; $T_{load} \gtrsim 10\,T_1$	MAT_280: FT = 0.060 GPa (calibrated, batch A), FTSCL = 1.2, FRACEN not set, EPSCR = 0.1; no `*MAT_ADD_EROSION`	Final calibrated model

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